



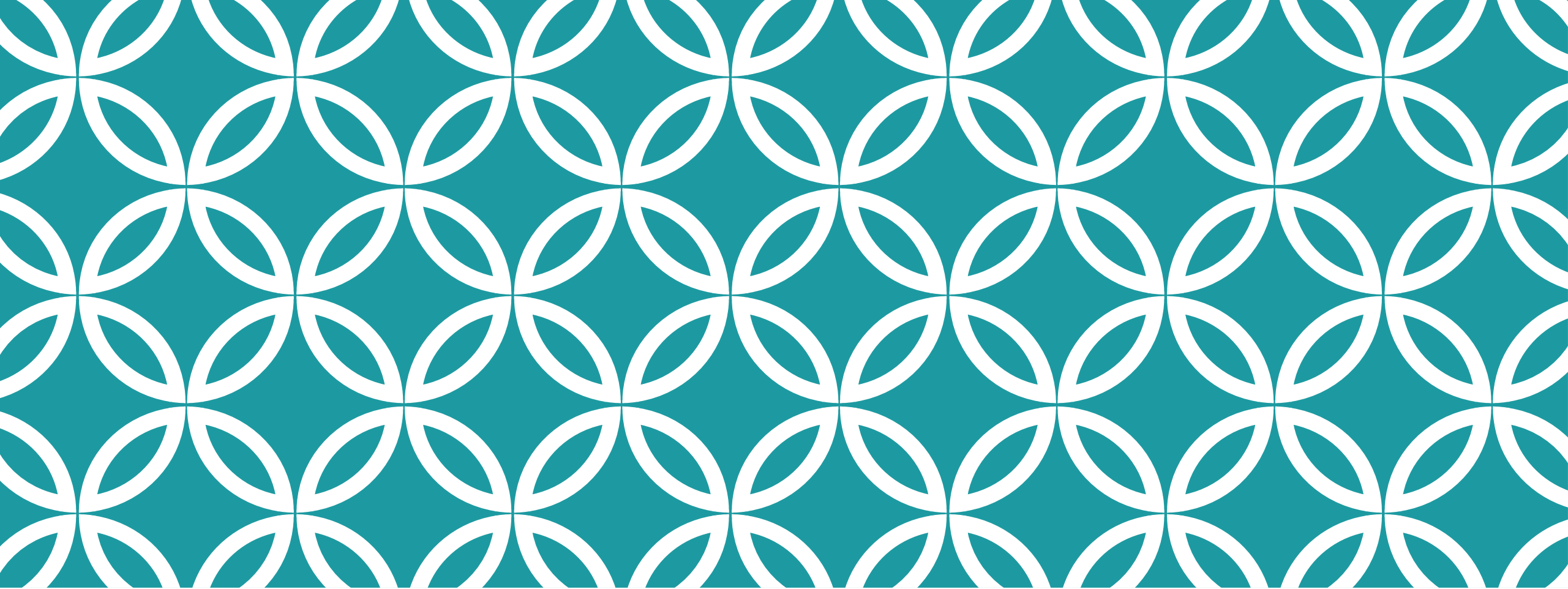
FLAME  
UNIVERSITY

# INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

PUBP 233

**Week 15: Demographic Dividend,  
Aging, and Future of Population**

Shivakumar Jolad

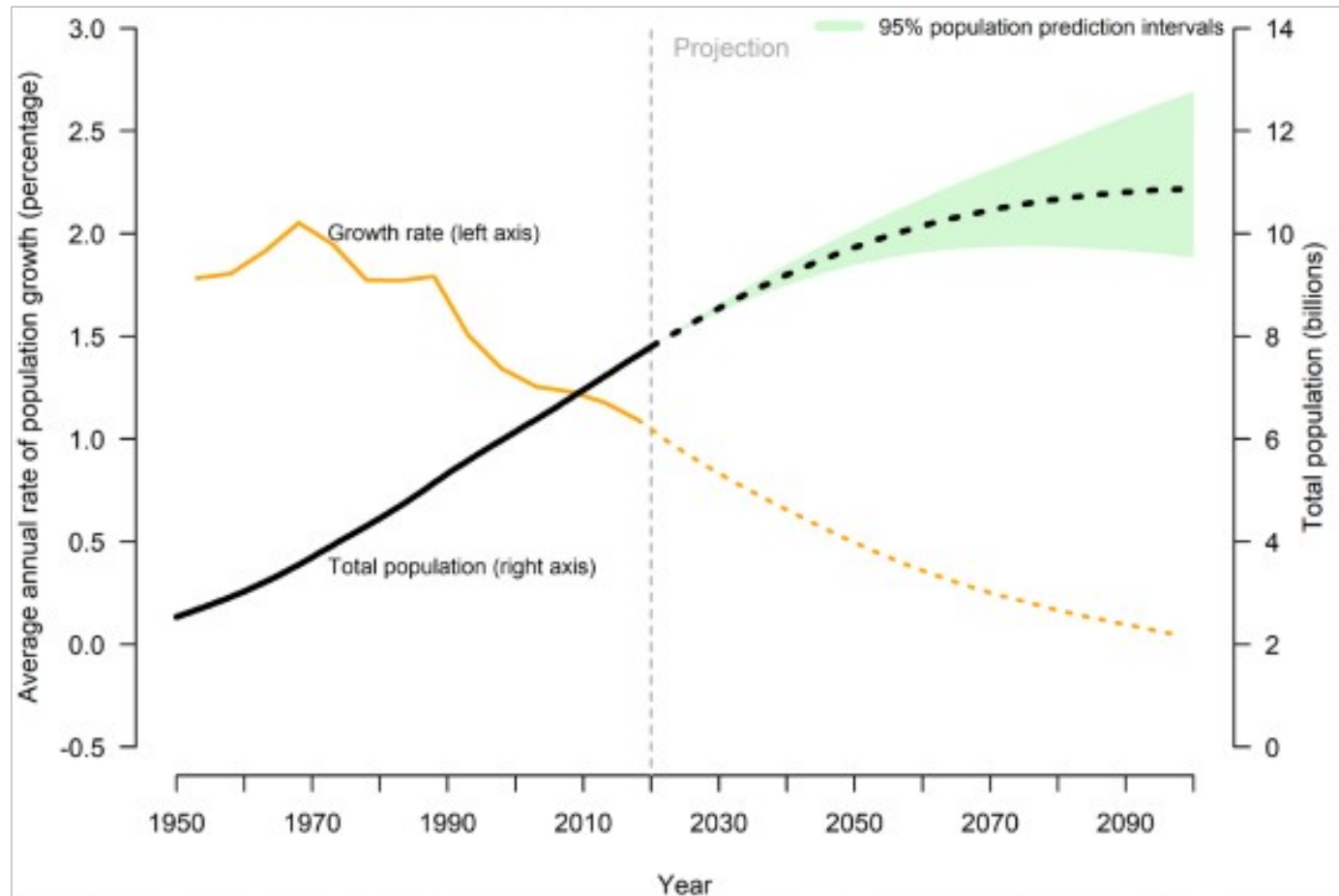


# FUTURE OF POPULATION |

# WORLD POPULATION AND GROWTH RATE

Figure 1. Population size and annual growth rate for the world: estimates, 1950-2020, and medium-variant projection with 95 per cent prediction intervals, 2020-2100

*Population growth continues at the global level, but the rate of increase is slowing, and the world's population could cease to grow around the end of the century*



Data source: United Nations, Department of Economic and Social Affairs, Population Division (2019). *World Population Prospects 2019*.

**Table 1. Population of the world, SDG regions and selected groups of countries, 2019, 2030, 2050 and 2100, according to the medium-variant projection**

| Region                           | Population (millions) |              |              |               |
|----------------------------------|-----------------------|--------------|--------------|---------------|
|                                  | 2019                  | 2030         | 2050         | 2100          |
| <b>World</b>                     | <b>7 713</b>          | <b>8 548</b> | <b>9 735</b> | <b>10 875</b> |
| Sub-Saharan Africa               | 1 066                 | 1 400        | 2 118        | 3 775         |
| Northern Africa and Western Asia | 517                   | 609          | 754          | 924           |
| Central and Southern Asia        | 1 991                 | 2 227        | 2 496        | 2 334         |
| Eastern and South-Eastern Asia   | 2 335                 | 2 427        | 2 411        | 1 967         |
| Latin America and the Caribbean  | 648                   | 706          | 762          | 680           |
| Australia/New Zealand            | 30                    | 33           | 38           | 49            |
| Oceania*                         | 12                    | 15           | 19           | 26            |
| Europe and Northern America      | 1 114                 | 1 132        | 1 136        | 1 120         |
|                                  |                       |              |              |               |
| Least developed countries        | 1 033                 | 1 314        | 1 877        | 3 047         |
| Land-locked Developing Countries | 521                   | 659          | 926          | 1 406         |
| Small Island Developing States   | 71                    | 78           | 87           | 88            |

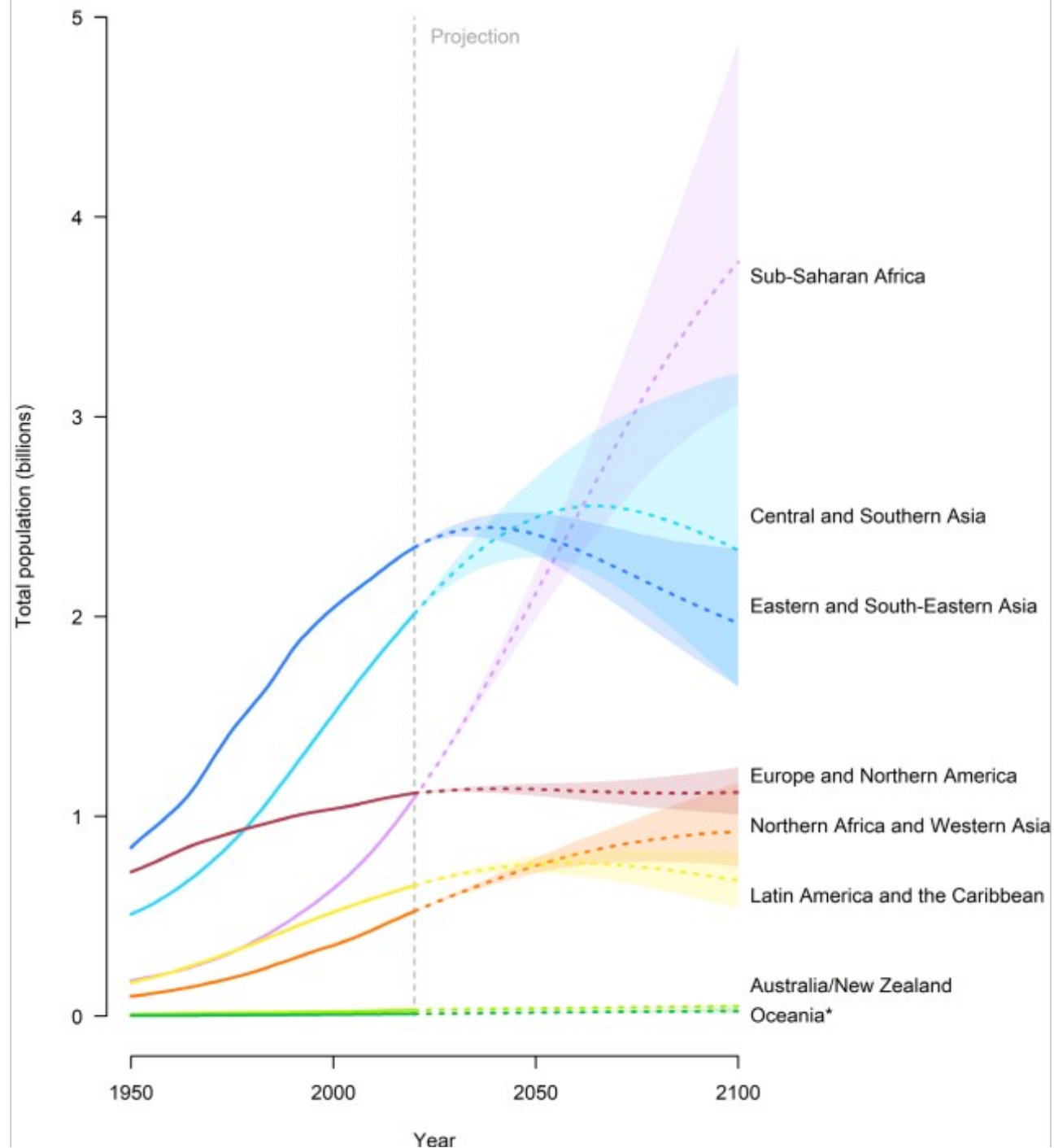
Data source: United Nations, Department of Economic and Social Affairs, Population Division (2019). *World Population Prospects 2019*.

\* excluding Australia and New Zealand

## World Population Prospects 2019

### Highlights





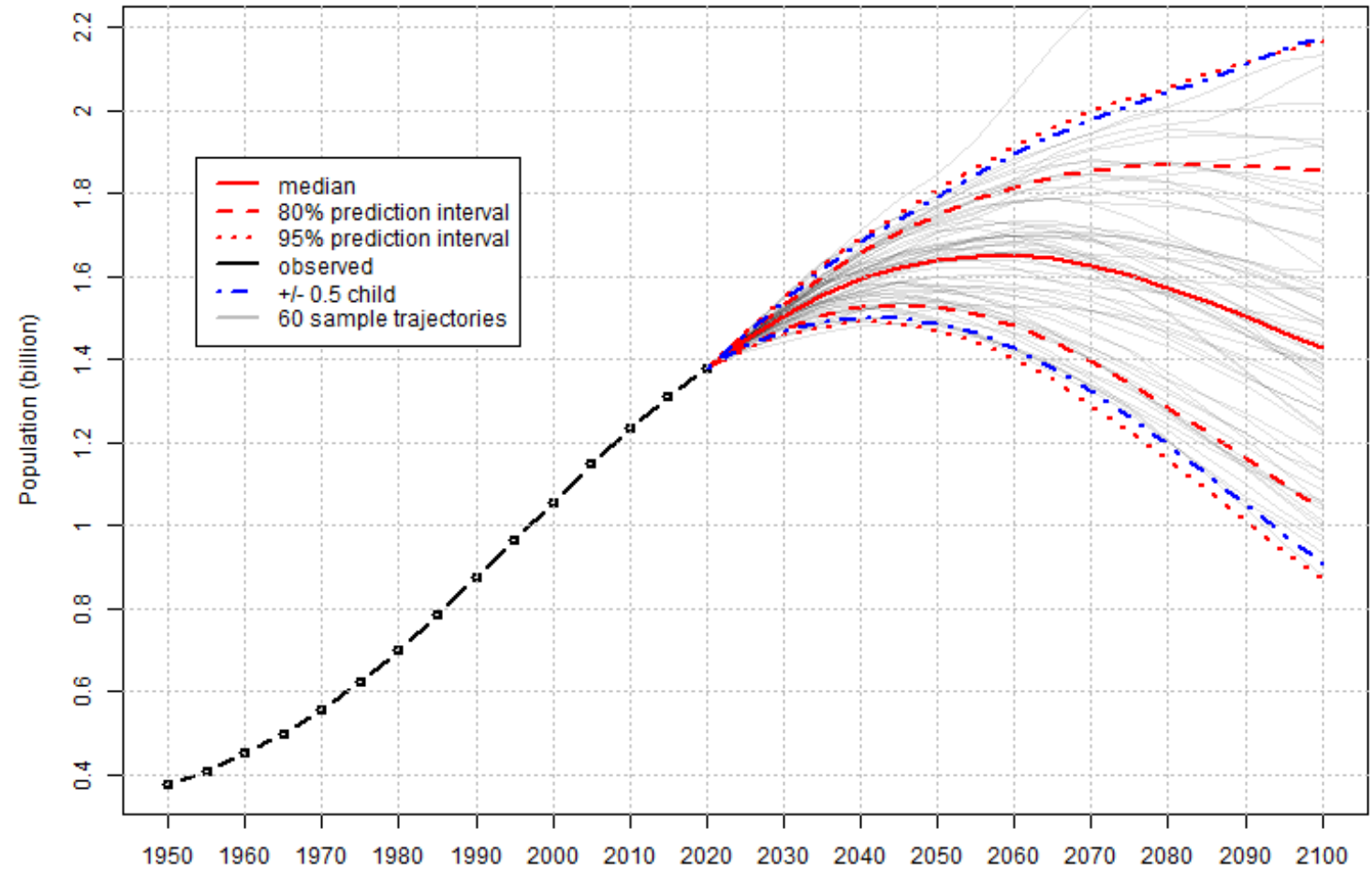
**Figure 2. Population by SDG region: estimates, 1950-2020, and medium-variant projection with 95 per cent prediction intervals, 2020-2100**

*Of the eight SDG regions, only sub-Saharan Africa is projected to sustain rapid population growth through the end of the century, according to the medium-variant projection*

## World Population Prospects 2019

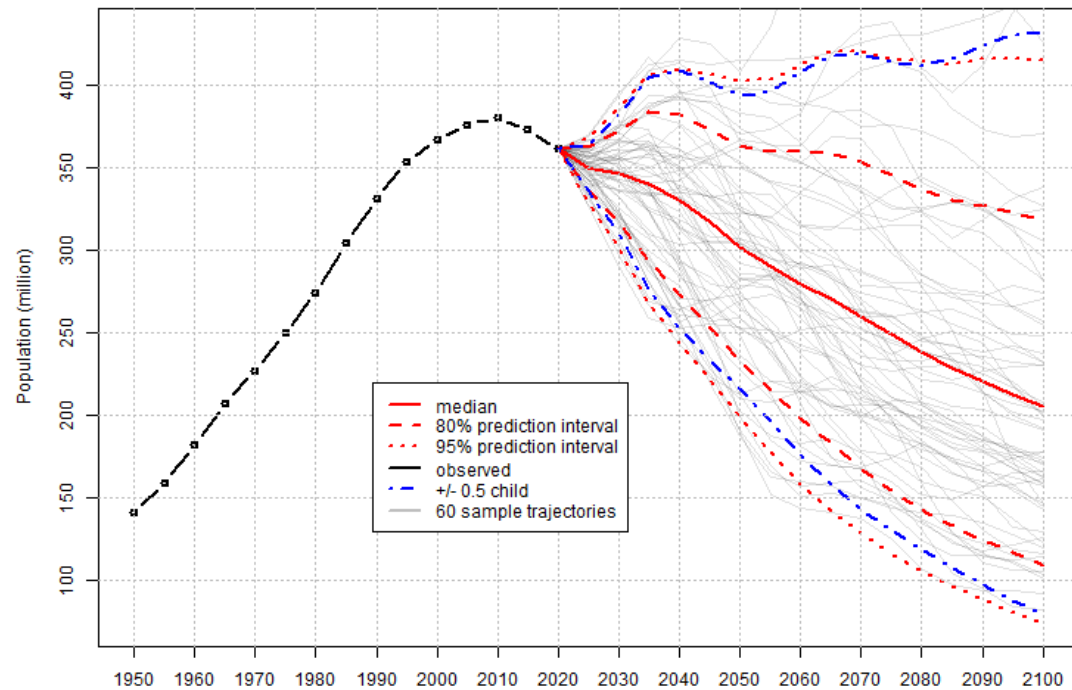
### Highlights

India: Total Population



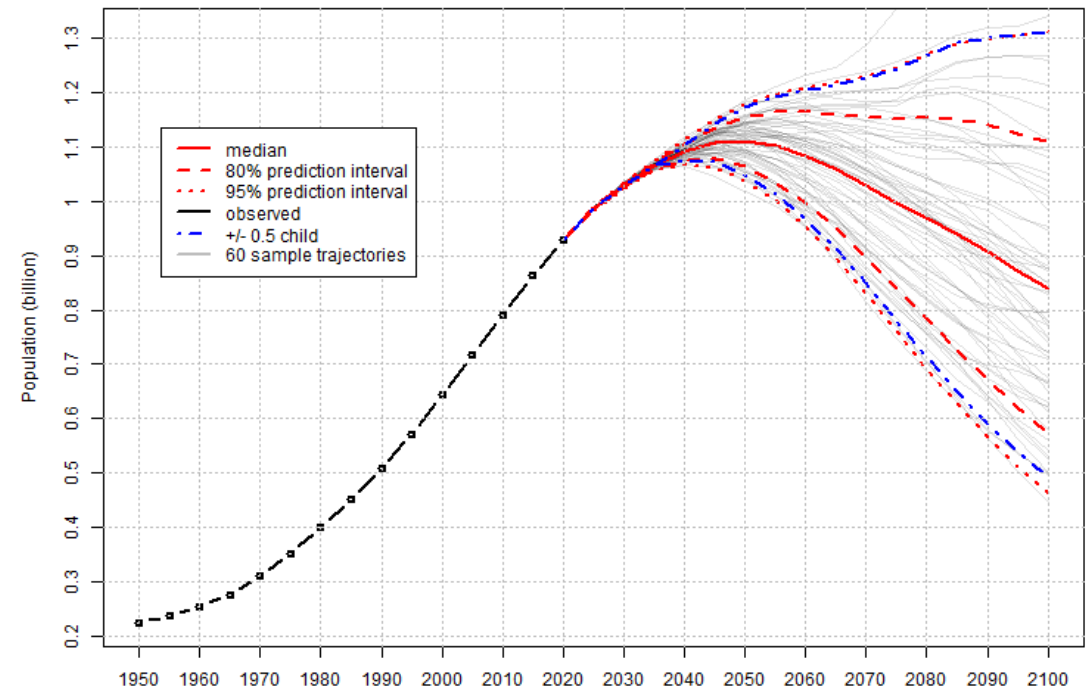
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United Nations, DESA, Population Division. *World Population Prospects 2019*. <http://population.un.org/wpp/>

India: Population (Age 0-14)



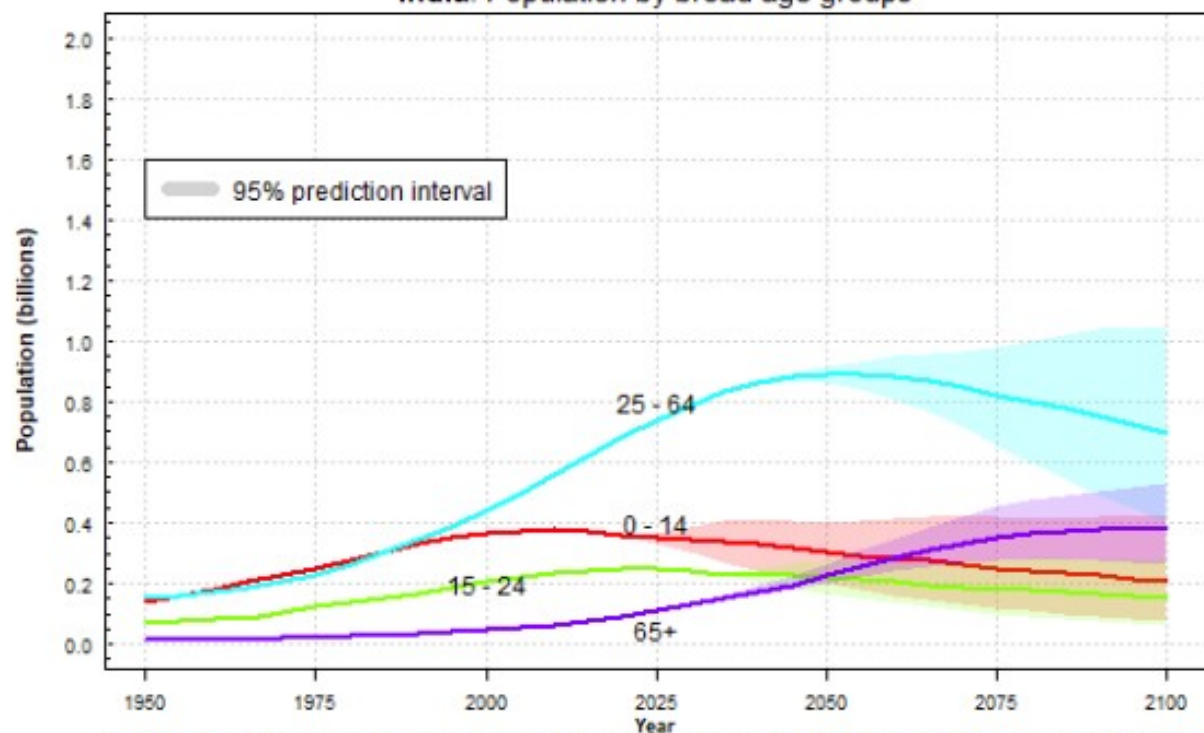
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United Nations, DESA, Population Division. *World Population Prospects 2019*. <http://population.un.org/wpp/>

India: Population (Age 15-64)



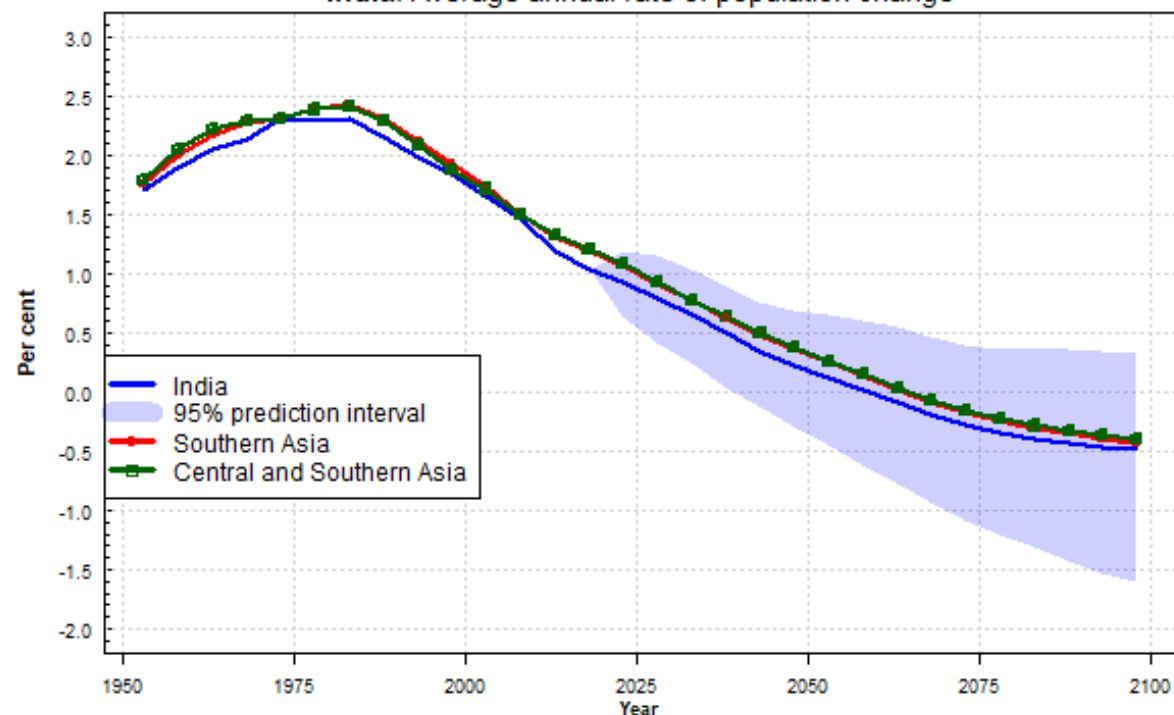
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United Nations, DESA, Population Division. *World Population Prospects 2019*. <http://population.un.org/wpp/>

**India: Population by broad age groups**



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United Nations, DESA, Population Division. *World Population Prospects 2019*. <http://population.un.org/wpp/>

**India: Average annual rate of population change**



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United Nations, DESA, Population Division. *World Population Prospects 2019*. <http://population.un.org/wpp/>



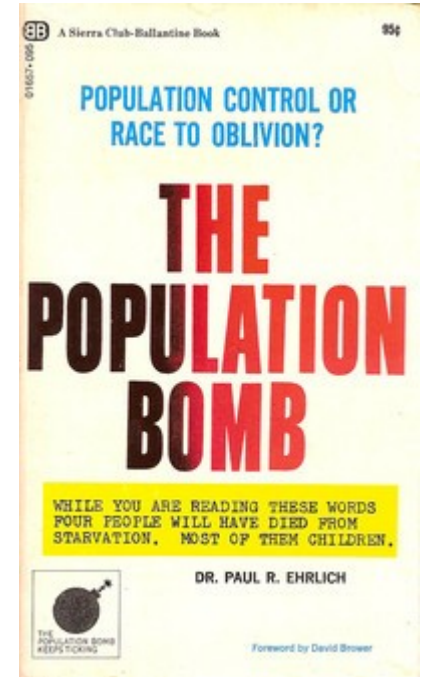


# POPULATION GROWTH, DEMOGRAPHIC DIVIDEND OR DISASTER?

# THE POPULATION BOMB

***"The battle to feed all of humanity is over. In the 1970s hundreds of millions of people will starve to death in spite of any crash programs embarked upon now. At this late date nothing can prevent a substantial increase in the world death rate."***

*"We must rapidly bring the world population under control, reducing the growth rate to zero or making it negative. Conscious regulation of human numbers must be achieved. Simultaneously we must, at least temporarily, greatly increase our food production."*



Paul Ralph Ehrlich,  
1968

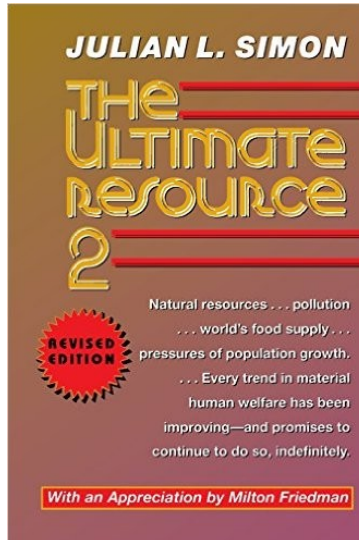
# REALITY CHECK

*Global death rate has continued to decline substantially since then, from 13/1000 in 1965–74 to 10/1000 from 1985–1990.*

The population of the world has more than doubled, while ***calories consumed/person have increased 24%.***



Julian  
Simon



*“Human beings create more than they destroy.”*

*“There is only one important resource which has shown a trend of increasing scarcity rather than increasing abundance. That resource is the most important of all—human beings. . . . [An] increase in the price of peoples’ services is a clear indication that people are becoming more scarce even though there are more of us.”*

– Simon, *The Ultimate Resource 2*, 580-581.

Simon–Ehrlich wager. (betting on whether the price of metals would decrease in 10 years)

“Adding more people causes problems. But people are also the means to solve these problems. The main fuel to speed the world’s progress is our stock of knowledge; the brakes are our lack of imagination and unsound social regulations of these activities. **The ultimate resource is people—especially skilled, spirited, and hopeful young people endowed with liberty—who will exert their wills and imaginations for their own benefits, and so inevitably they will benefit the rest of us as well.**”

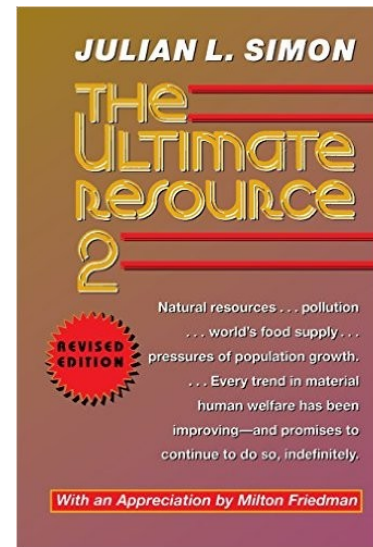
– Julian Simon, “Introduction,” *The State of Humanity*, 27

“The world’s problem is not too many people, **but a lack of political and economic freedom.**”

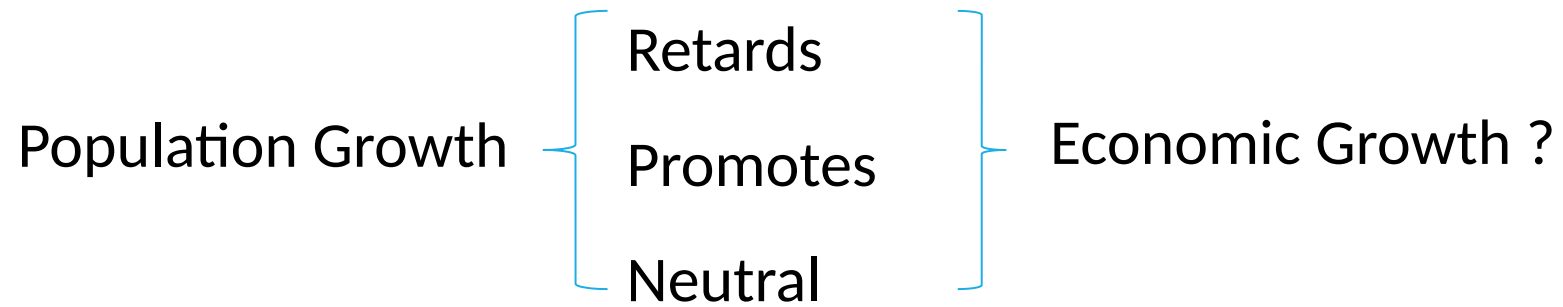
– Julian Simon, *The Ultimate Resource 2* (Princeton, N.Y.: Princeton University Press, 1996), p. 11.

“Discoveries, like resources, may well be infinite: the more we discover, the more we are able to discover.”

– Simon, *The Ultimate Resource 2*, p. 82.



# POPULATION: RESOURCE OR LIABILITY?



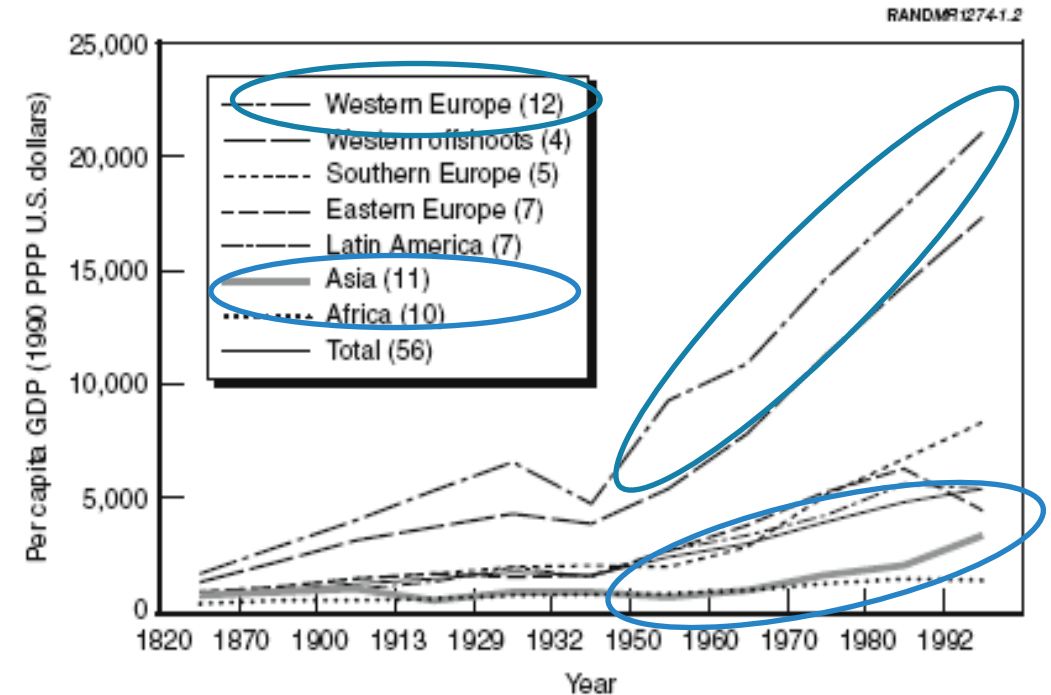


# PESSIMISTIC VIEW

Thomas Malthus (1798)  
Paul Ehrlich (1968)

High population greater need for resources

Large Investment needed to supply needs for growing population than increase provision per capita



SOURCE: Angus Maddison, *Monitoring the World Economy: 1820–1992*, Paris: OECD, 1995.

Kent, Mary and Haub, Carl. 2005. *Global Demographic Divide*  
Population Bulletin: 60 (4)

Bloom, David E., Canning, David. and Sevilla, Jaypee, 2002. **Demographic Dividend- the Economic consequence of population change**, Population matters, RAND Corporation.

# OPTIMISTIC VIEW

Millions People didn't starve due to population growth :  
***Green Revolution* quadrupled food production**

Per capita income didn't decrease with rapid population

Drivers of Economic growth:

- **Human capital** and **technology** (than physical capital )
- Social and Institutional Innovation
- **Knowledge driven sectors**

Kuznets and Simon:

*Population increase, so does human ingenuity. Large societies can better exploit economy of scale and human innovation.*

*"The Ultimate Resource"* (Julian Simon, 1987)

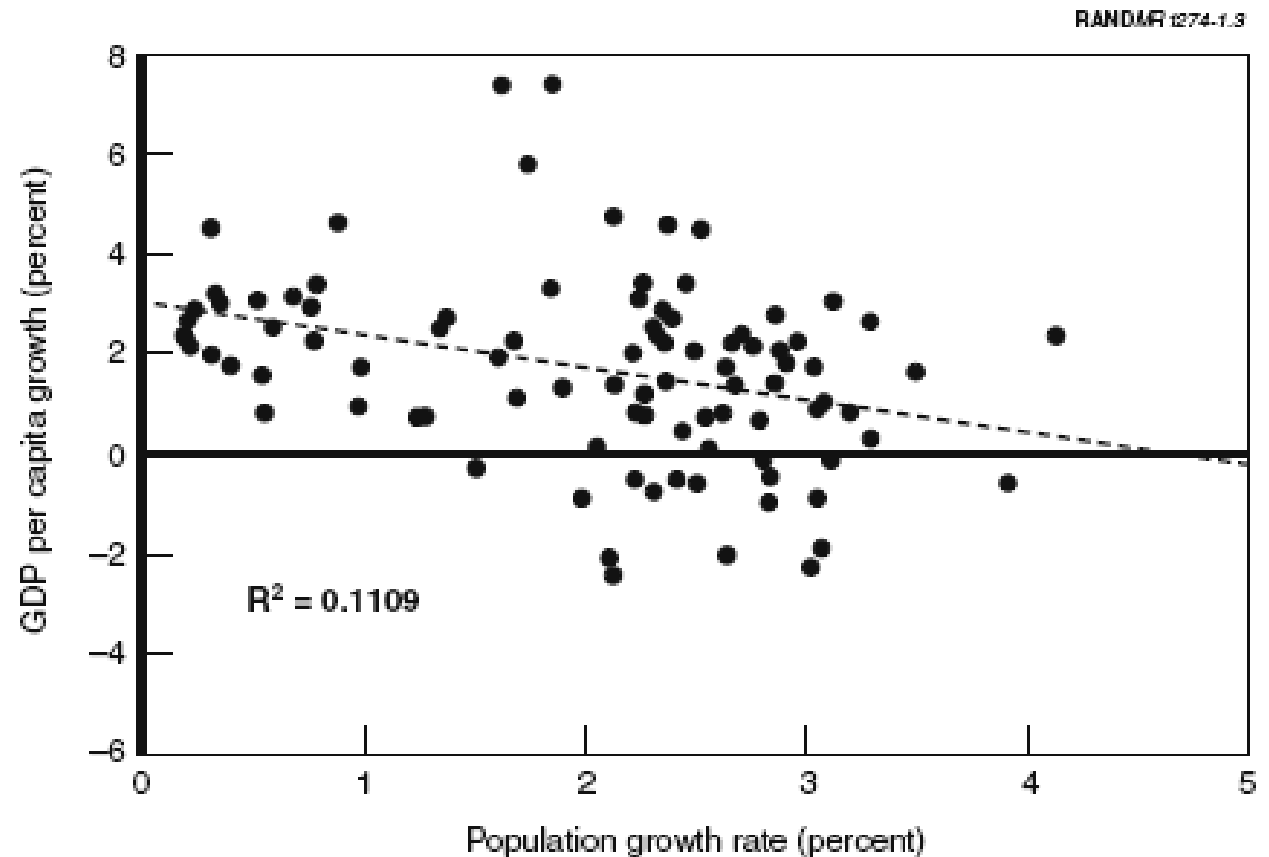
# NEUTRALIST VIEW

Economic growth and population growth shows little statistically significant correlation

Little negative correlation disappears

- **Educational attainment**
- Openness to trade
- Country size
- Civil and Political institutions

*“On balance slow population growth is beneficial to most developing nations” (NRC,1986)*



# FERTILITY DECLINE AND CHANGE IN AGE STRUCTURE

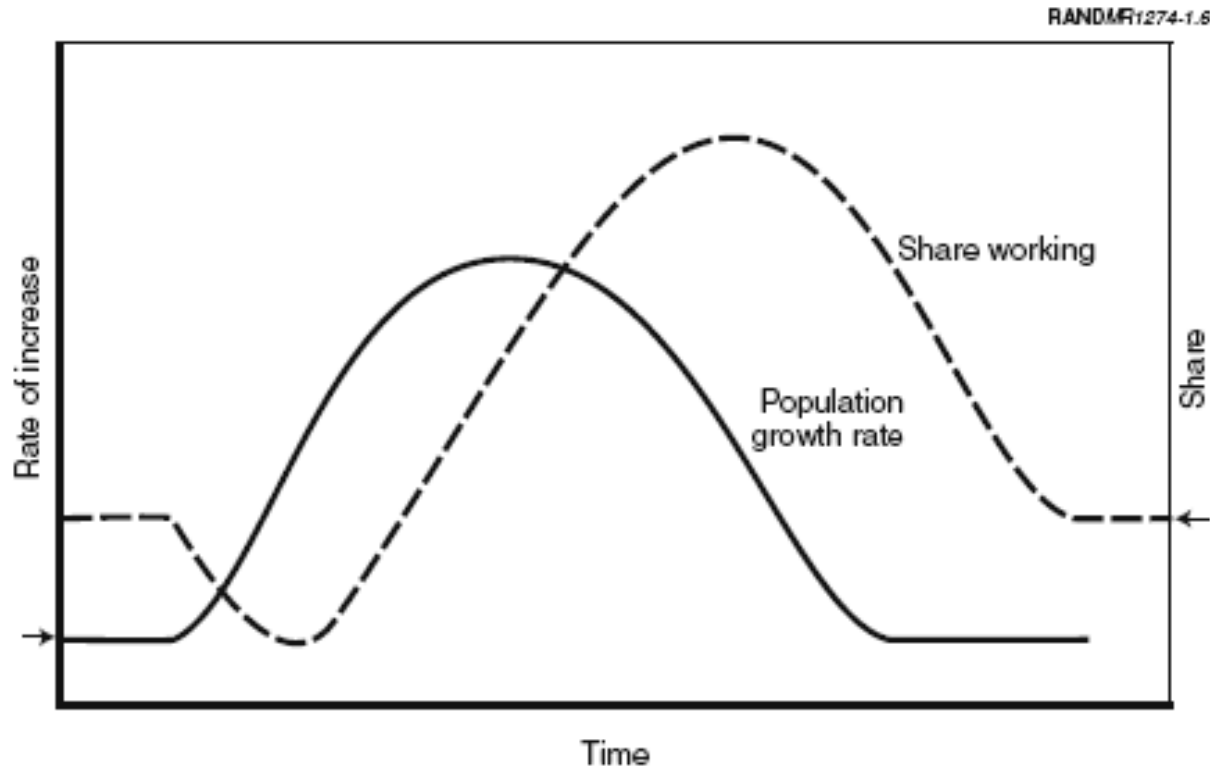
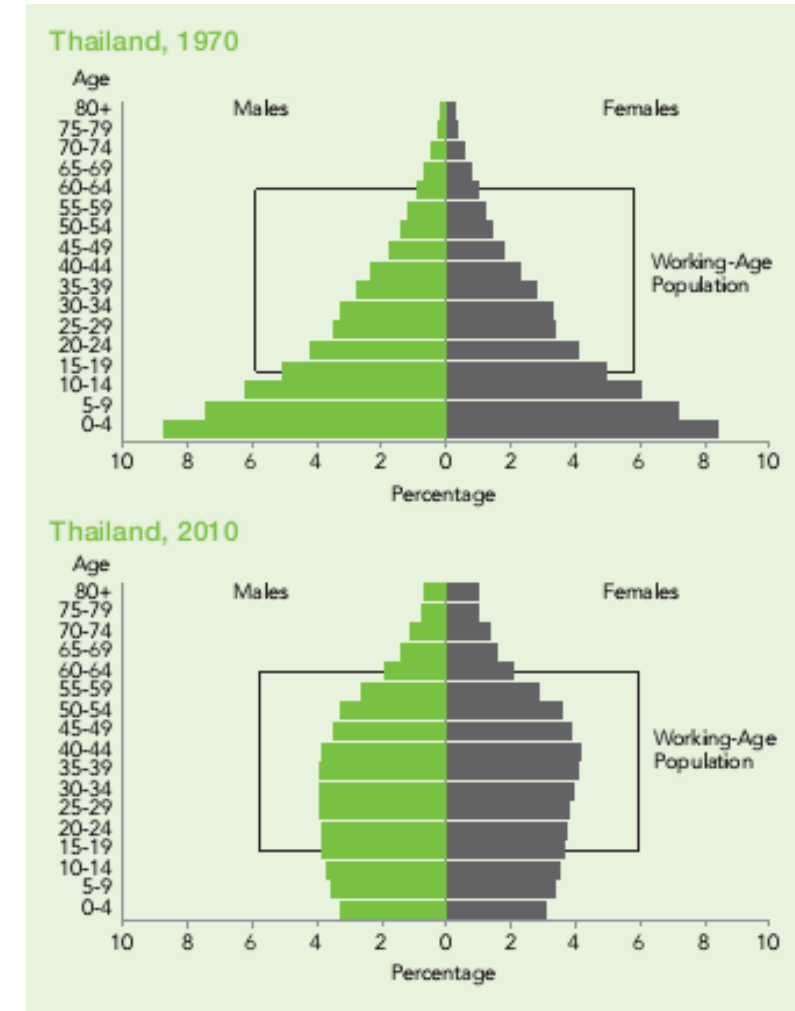


Figure 1.6—Population Growth and Age Structure

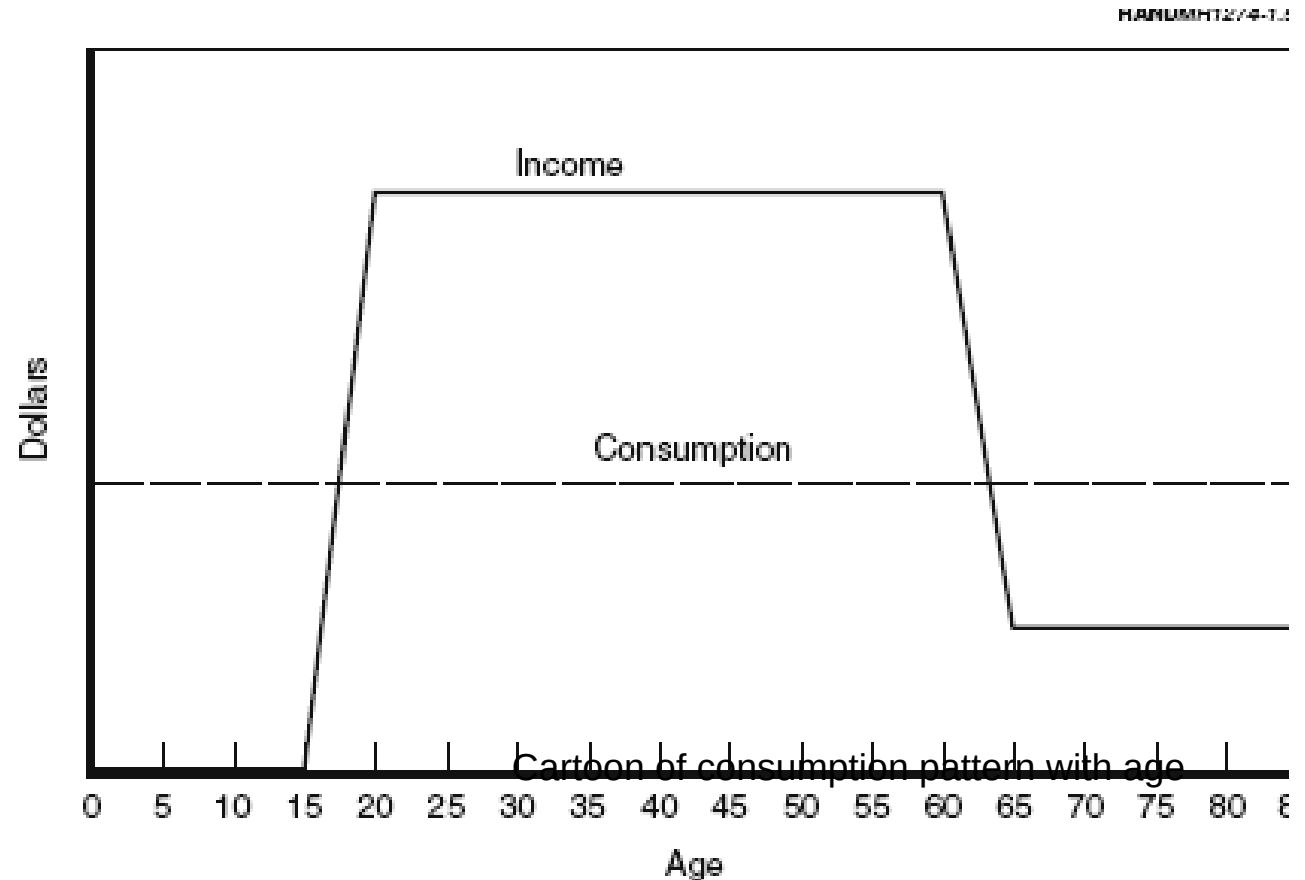


# DEMOGRAPHIC DIVIDEND

The demographic dividend refers to the **accelerated economic growth that begins with changes in the age structure** of a country's population as it transitions from high to low birth and death rates.

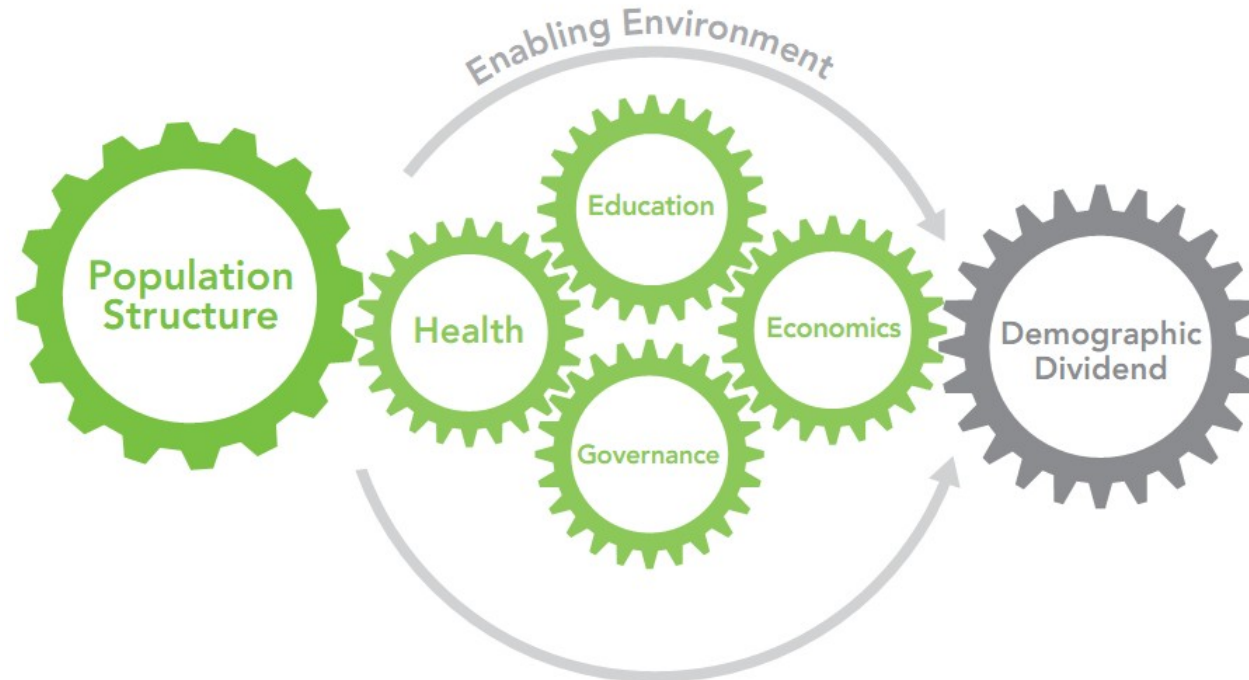
When dependency ratio is minimum

*'Demographic bonus'*



# POLICY INTERVENTIONS FOR REAPING DEMOGRAPHIC DIVIDEND

Right policy environment needed for Capitalizing the demographic dividend



Absence of good policies, unemployment, and socio-political instability puts strain on health, education and social welfare systems



# EAST ASIA: ASIAN TIGERS

Fastest demographic transition to date

High (6) to replacement level TFR in just 30 years; Working age grew 4 x dependent population

Demographic Dividend explains 1/3<sup>rd</sup> of the *Economic Miracle* (Asian Tigers)

Investment in Education, rapid expansion of labor market facilitated the growth

Figure 6. Stylized Model of Economic Growth and the Demographic Transition in East Asia, 1945–2025

Growth rate of real GDP per capita

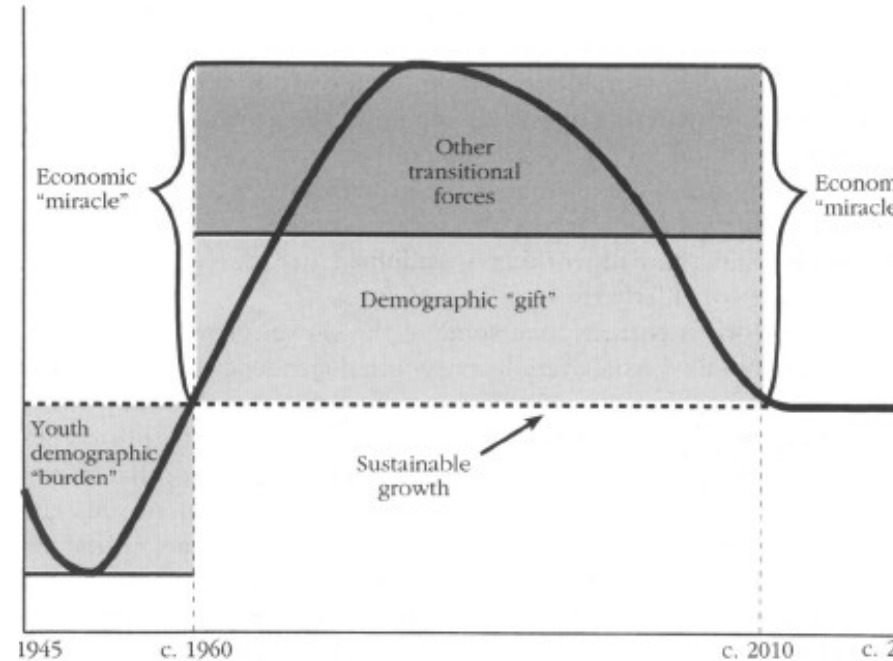
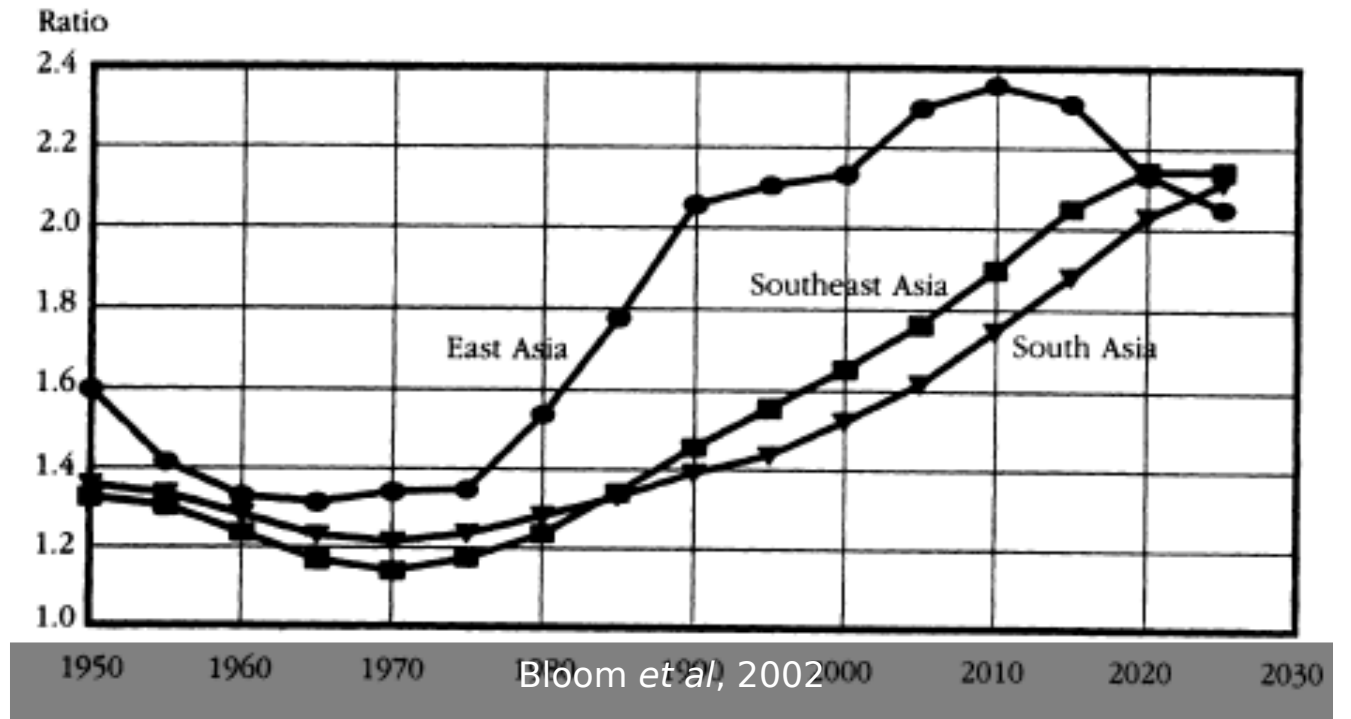
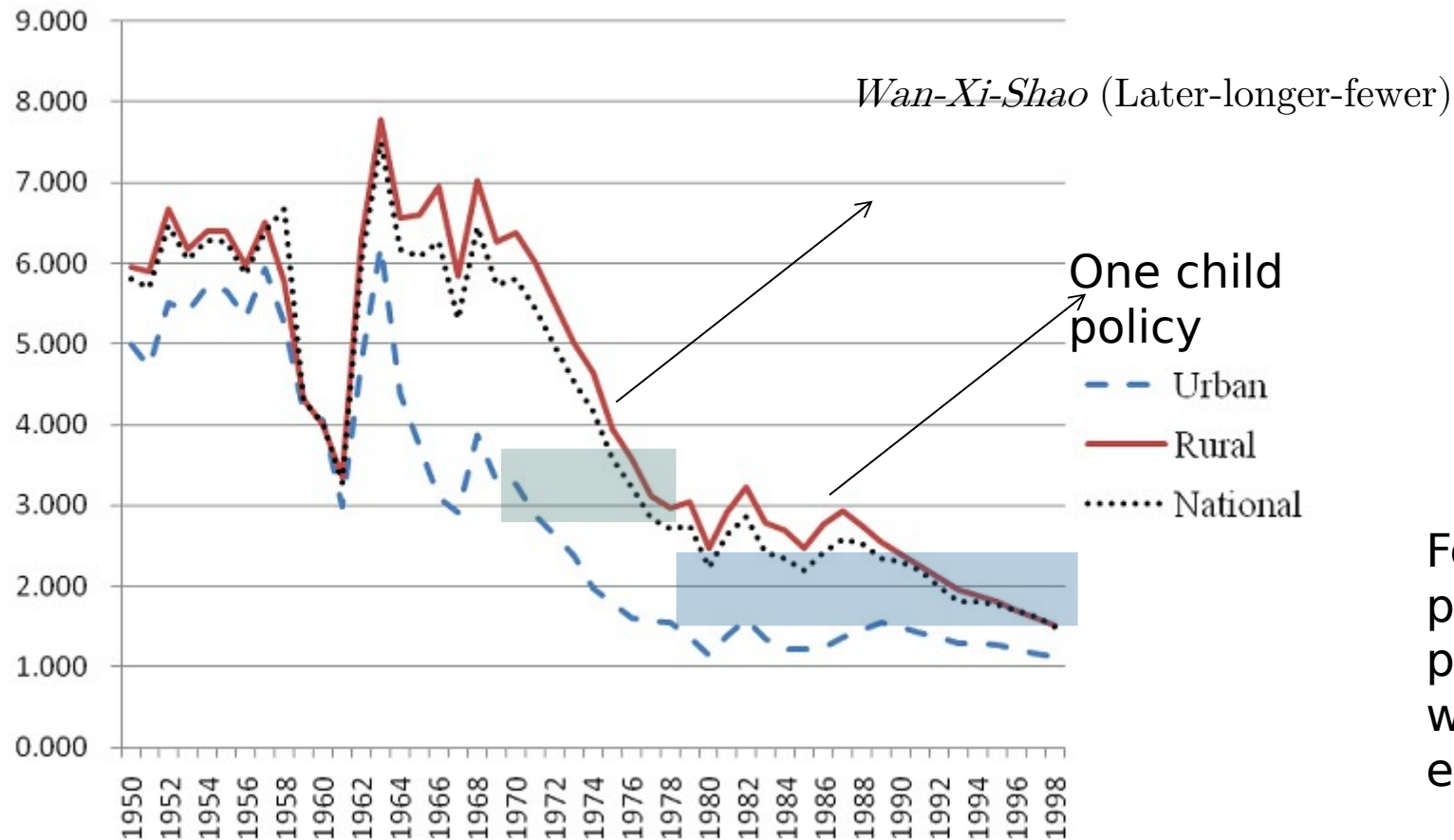


Figure 5. Ratio of Working-age to Nonworking-age Population in Asia, 1950–2030



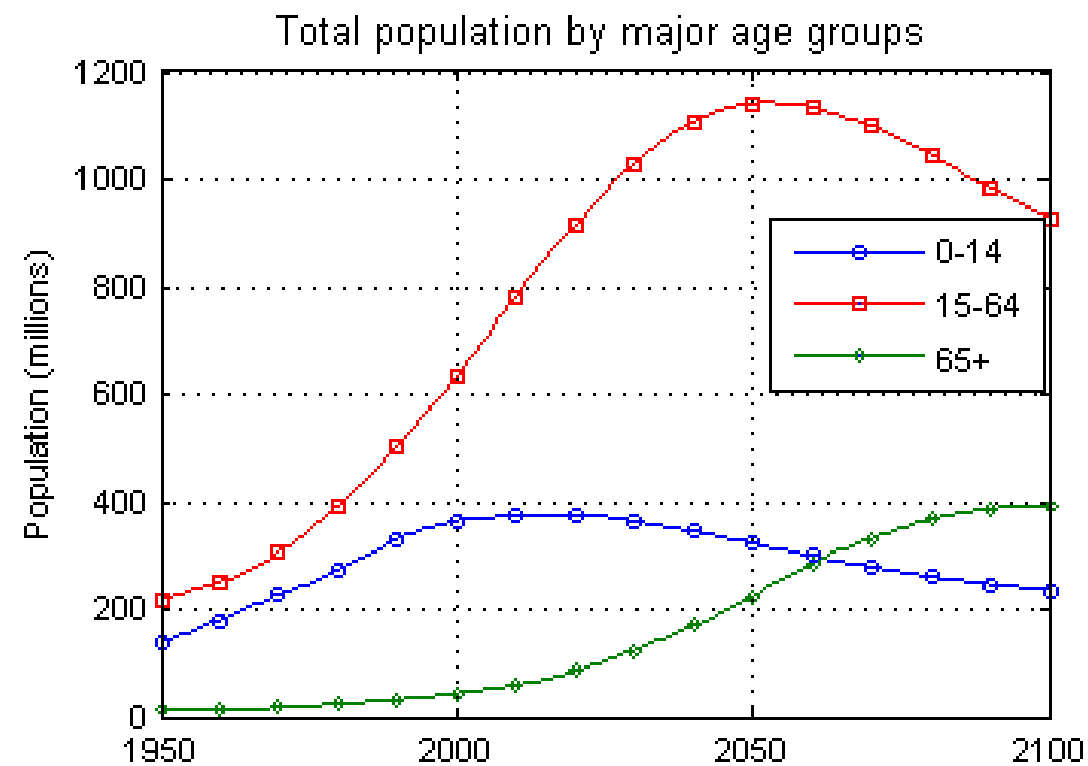
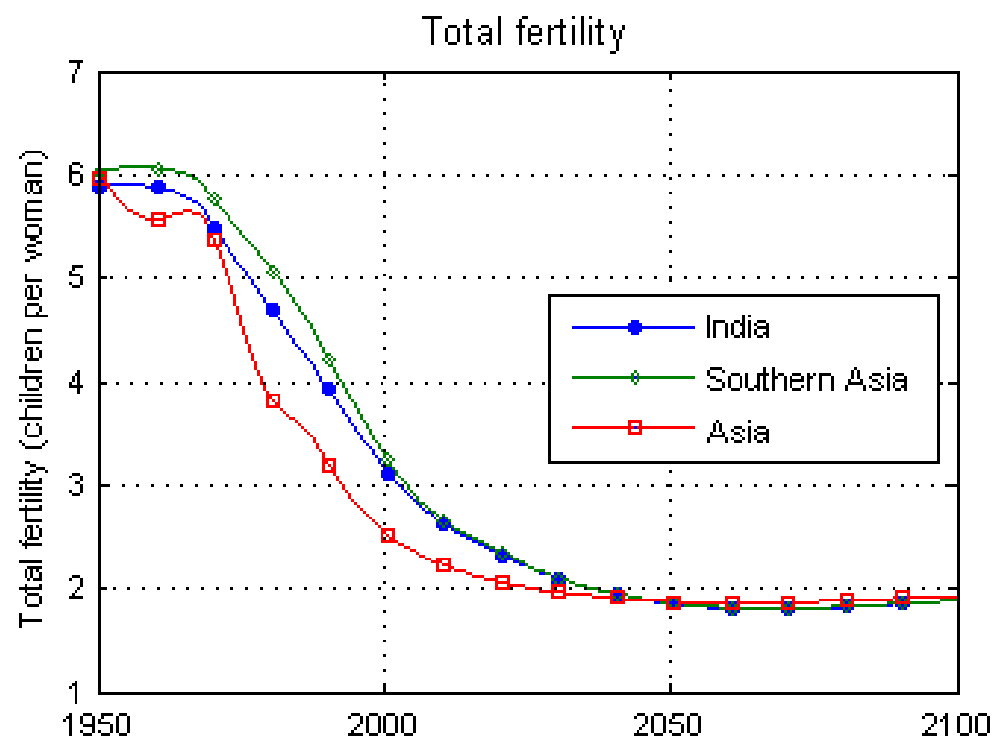
# FERTILITY DECLINE- NATURAL V/S FORCED

Figure 1: China's Total Fertility Rate, 1950-1998



Female labor force participation (1982) 85.3 percent of Chinese women ages 15 to 54 employed

Wu, Xiaogang, Hua Ye, and Guangye He. "Fertility Decline and Women's Empowerment in China." (2012).



# DEMOGRAPHIC DIVIDEND OR DISASTER?

Bloomberg | Quint

Markets

Business

Global Economics

Budget 2017

Opinion



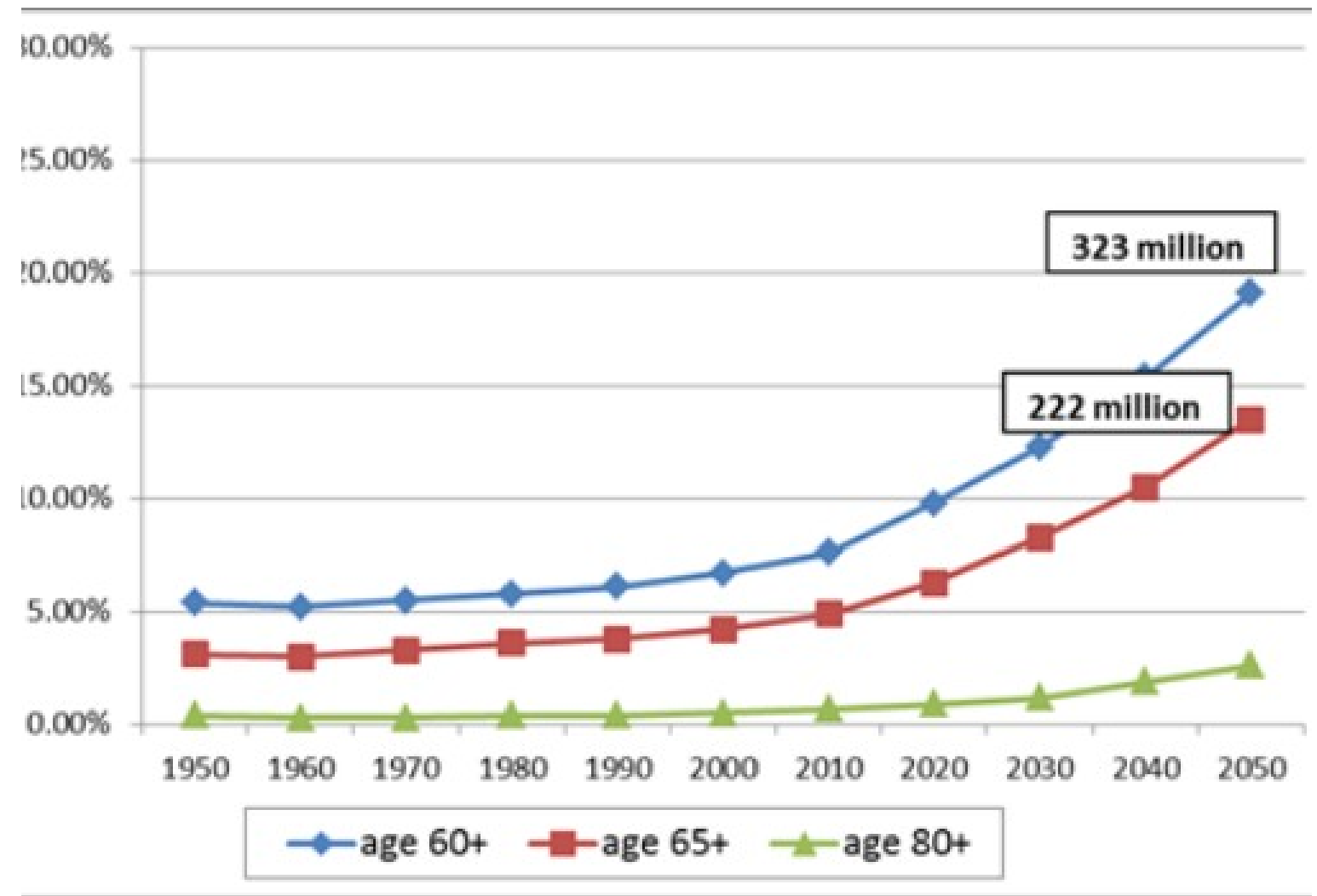
Villagers hold up their job cards with no entries for the rural jobs program. (Photograph: Prashanth Vishwanathan/Bloomberg)

## India Is In The Midst Of A Demographic Disaster: Mohandas Pai

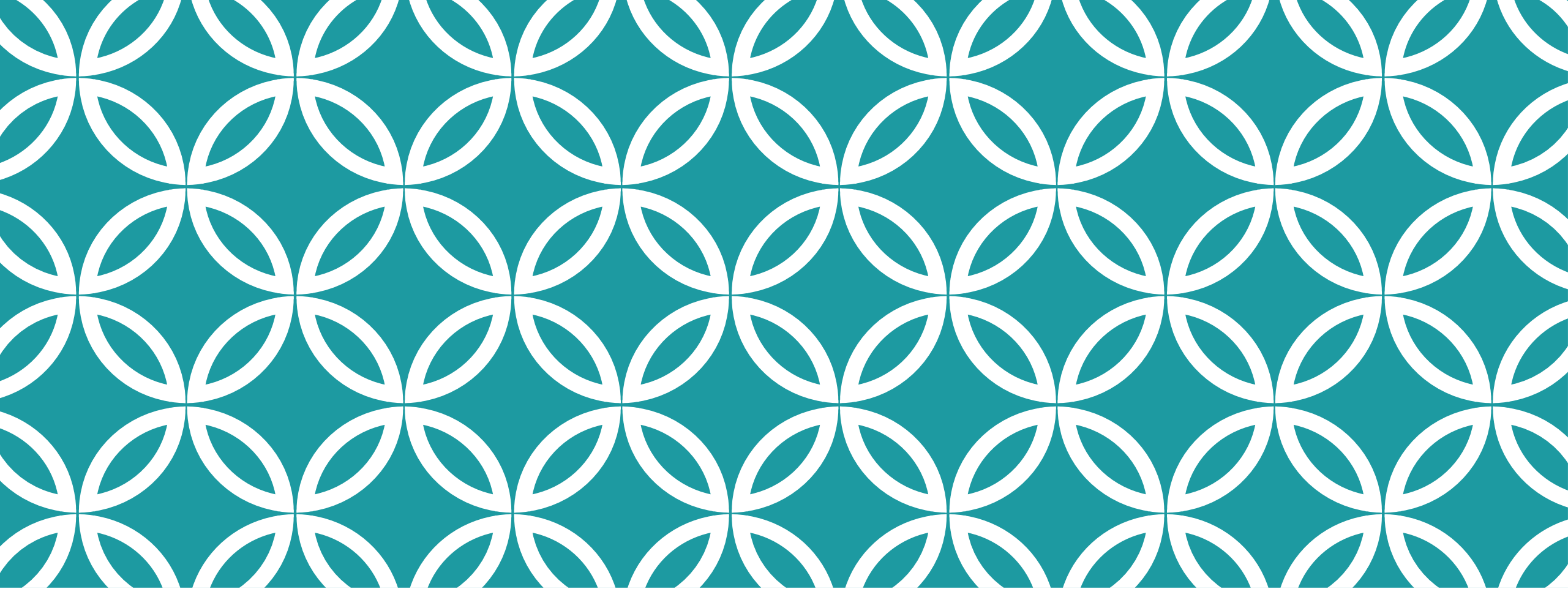
by Sajeet Manghat  
[@sajeetkm](#)

Updated on January 11, 2017, 11:04 am  
January 11, 2017, 11:04 am

Figure 1: India's changing age structure and population aging



Tannistha  
Samanta



# DEMOGRAPHIC TRANSITION AND URBANIZATION



# URBANISATION AND GROWTH OF CITIES IN INDIA



<https://www.youtube.com/watch?>



## Urbanisation, demographic transition, and the growth of cities in India, 1870-2020

Chinmay Tumbe, 2016



# URBANIZATION AND URBAN GROWTH

Urbanization and urban growth– Linked to unfolding **demographic transition** in India.

### Urban Growth and Urbanization

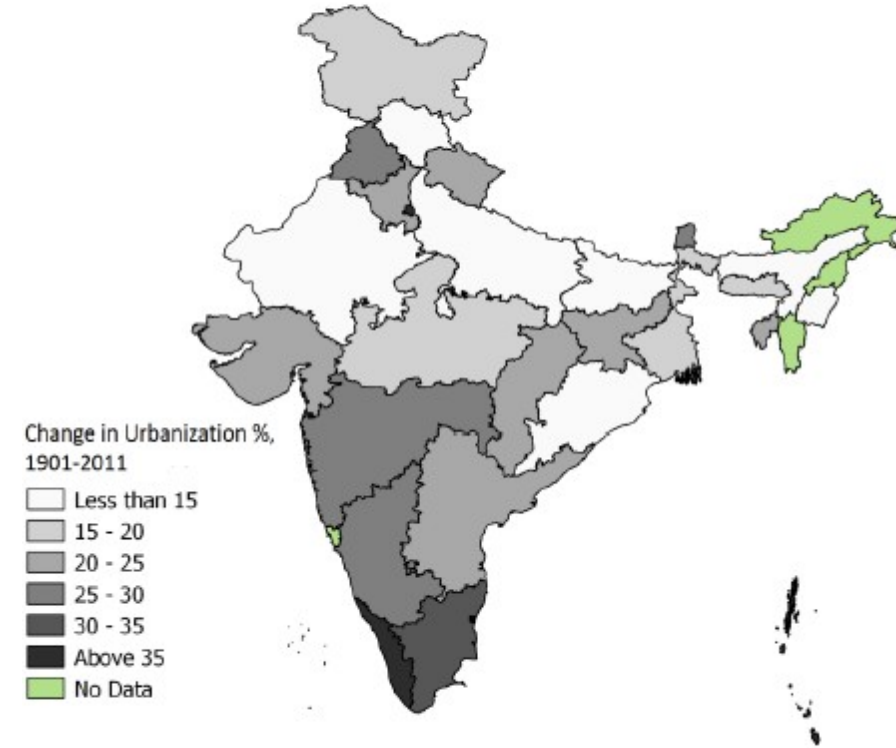
- ▢ Urbanization depends not only on urban growth but also on rural growth, a key point emphasized in this paper.
- ▢ The demographic transition, on the other hand, refers to the transition from high birth and death rates to low birth and death rates.

scenarios. The analysis itself is based on a unique database of demographic, economic, geographic and social indicators covering regions and cities over time. It assembles a database on the population of nearly 400 urban agglomerations (1871-2011) that collectively comprise of over 60% of the Indian urban population. Data on 4,000 urban units in 2001, district level metrics and a State level panel between 1971 and 2011 are also analysed. The Appendix to this paper provides details on the compilation of the database along with summary statistics of key variables.

**Table 1: Urbanization and Urban Growth in India, 1871-2011**

| Year | Number of Towns and Cities | Number of Cities/UAs with Million+ Population | Urban Population (Million) | Total Population (Million) | Urbanization % | Annual Inter-Censal Urban Population Growth Rate % | Annual Inter-Censal Total Population Growth Rate % |
|------|----------------------------|-----------------------------------------------|----------------------------|----------------------------|----------------|----------------------------------------------------|----------------------------------------------------|
|      | (1)                        | (2)                                           | (3)                        | (4)                        | (5)            | (6)                                                | (7)                                                |
| 1871 |                            | 1                                             | 18                         | 212                        | 8.7            |                                                    |                                                    |
| 1881 |                            | 1                                             | 20                         | 214                        | 9.3            | 0.75                                               | 0.08                                               |
| 1891 |                            | 1                                             | 22                         | 234                        | 9.4            | 1.03                                               | 0.92                                               |
| 1901 | 1,916                      | 1                                             | 26                         | 238                        | 10.8           | 1.63                                               | 0.19                                               |
| 1911 | 1,908                      | 2                                             | 26                         | 252                        | 10.3           | 0.04                                               | 0.56                                               |
| 1921 | 2,048                      | 2                                             | 28                         | 251                        | 11.2           | 0.80                                               | -0.03                                              |
| 1931 | 2,220                      | 2                                             | 33                         | 279                        | 12.0           | 1.77                                               | 1.05                                               |
| 1941 | 2,427                      | 2                                             | 44                         | 319                        | 13.9           | 2.81                                               | 1.34                                               |
| 1951 | 3,060                      | 5                                             | 62                         | 361                        | 17.3           | 3.52                                               | 1.26                                               |
| 1961 | 2,700                      | 7                                             | 79                         | 439                        | 18.0           | 2.37                                               | 1.98                                               |
| 1971 | 3,126                      | 9                                             | 109                        | 548                        | 19.9           | 3.29                                               | 2.24                                               |
| 1981 | 4,029                      | 12                                            | 159                        | 683                        | 23.3           | 3.87                                               | 2.23                                               |
| 1991 | 4,689                      | 23                                            | 218                        | 846                        | 25.7           | 3.16                                               | 2.16                                               |
| 2001 | 5,161                      | 35                                            | 286                        | 1,029                      | 27.8           | 2.78                                               | 1.97                                               |
| 2011 | 7,935                      | 53                                            | 377                        | 1,211                      | 31.2           | 2.80                                               | 1.64                                               |

**Figure 2: Change in State Urbanization %, 1901-2011**



# RURAL- URBAN SEX RATIOS OF DISTRICTS- 2011

Figure 3 (B): Rural-Urban Sex Ratios across Indian Districts

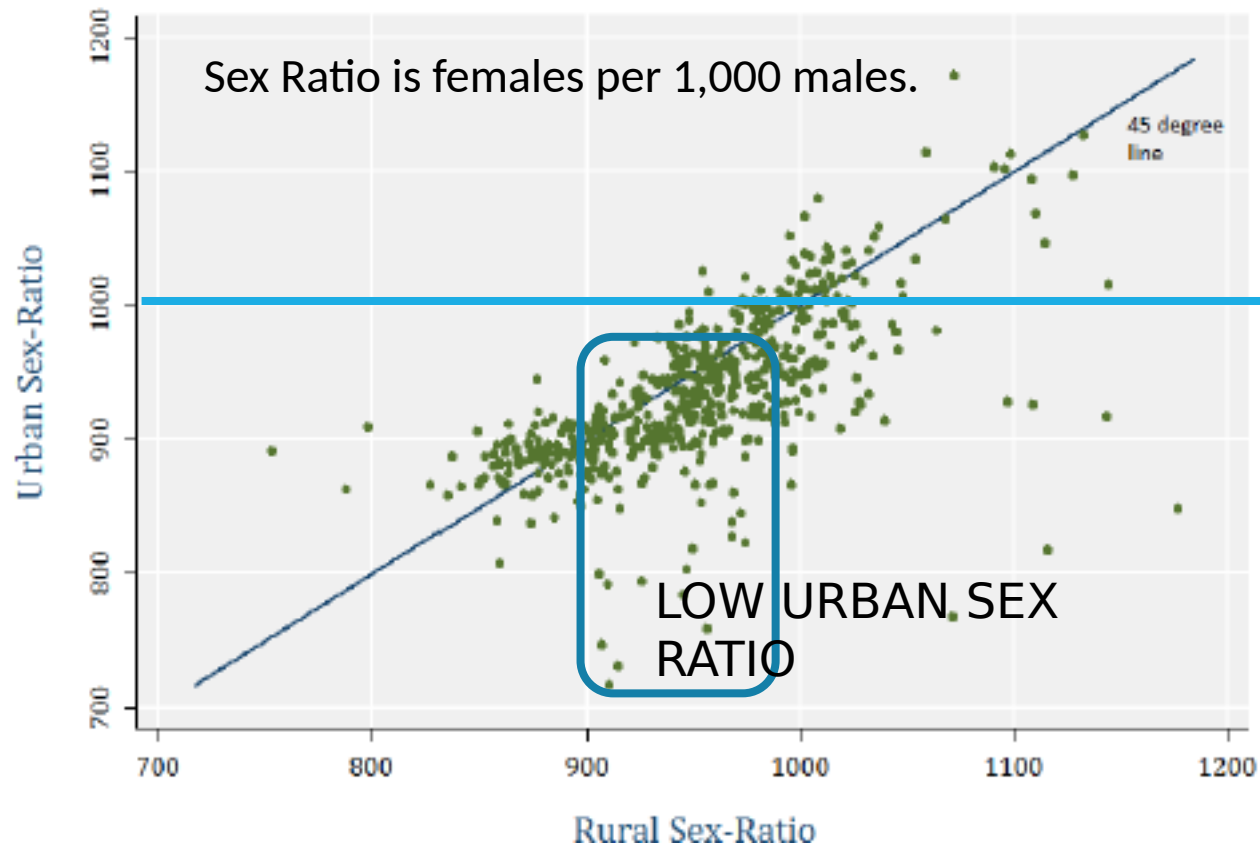
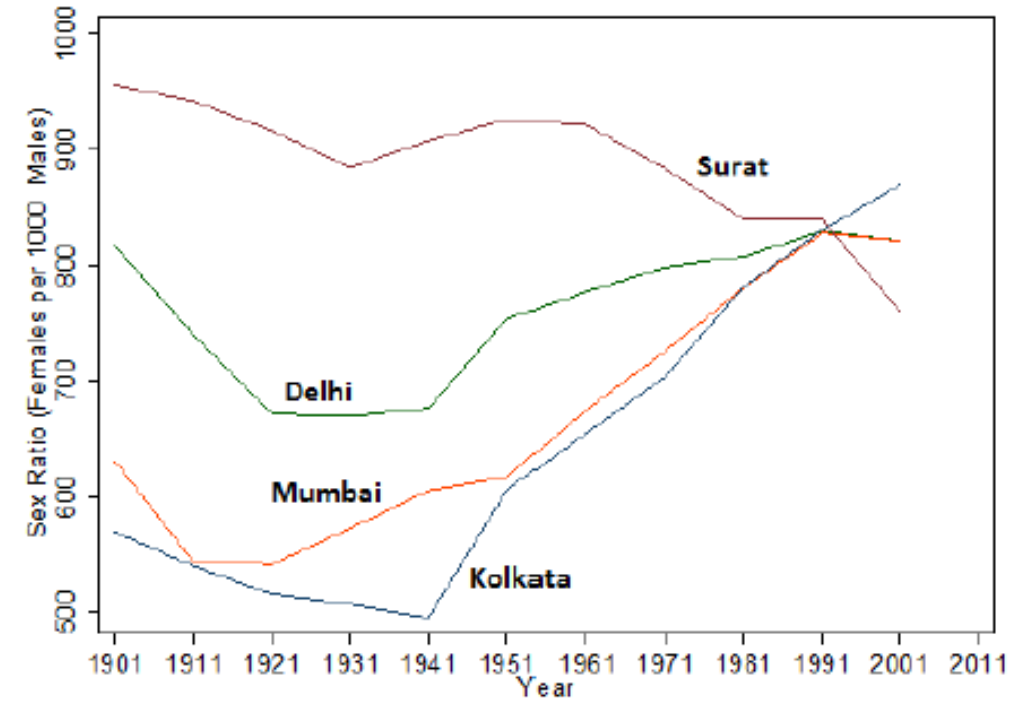


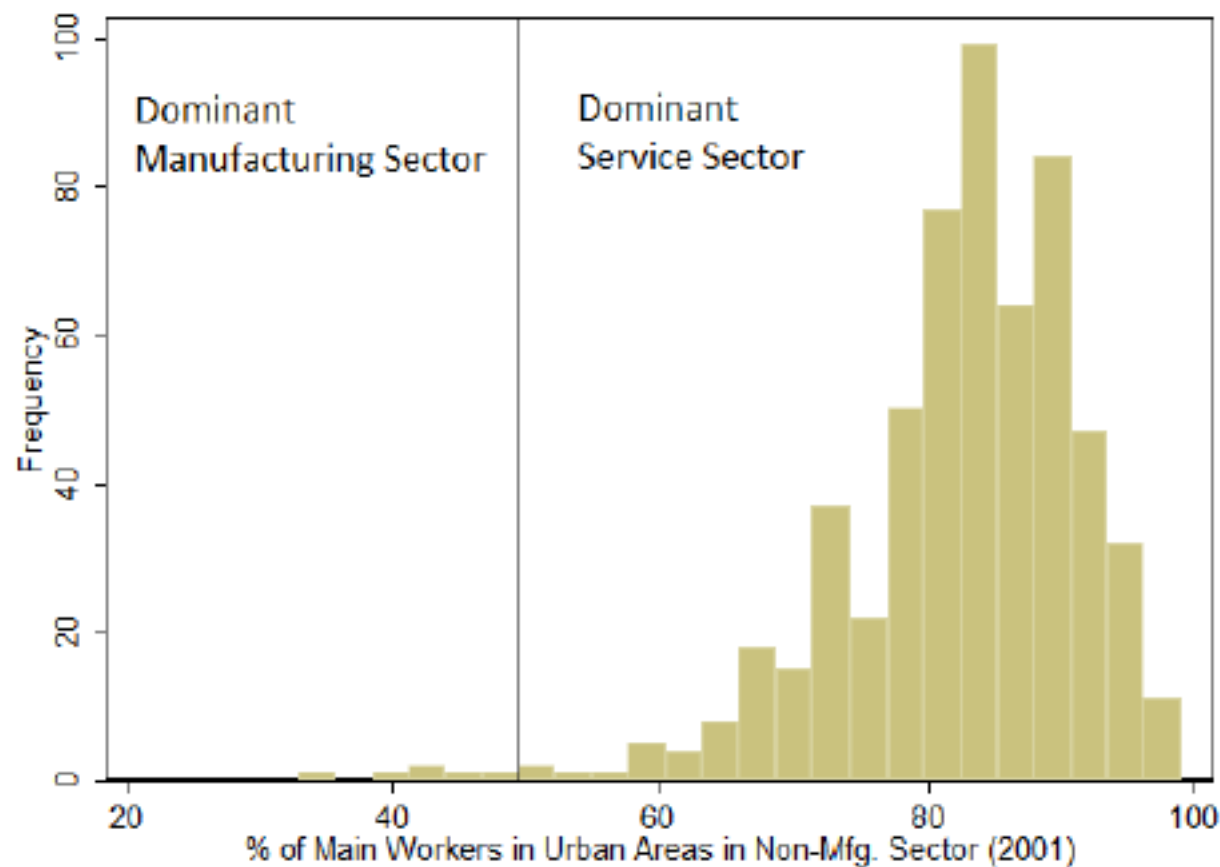
Figure 4: Selected City Sex Ratios, 1901-2011



Source: Census 1901-2011. Data on Urban Agglomerations.

# SERVICE SECTOR INDIANIZATION

Figure 5: District Level Distribution of Urban Workers, 2001



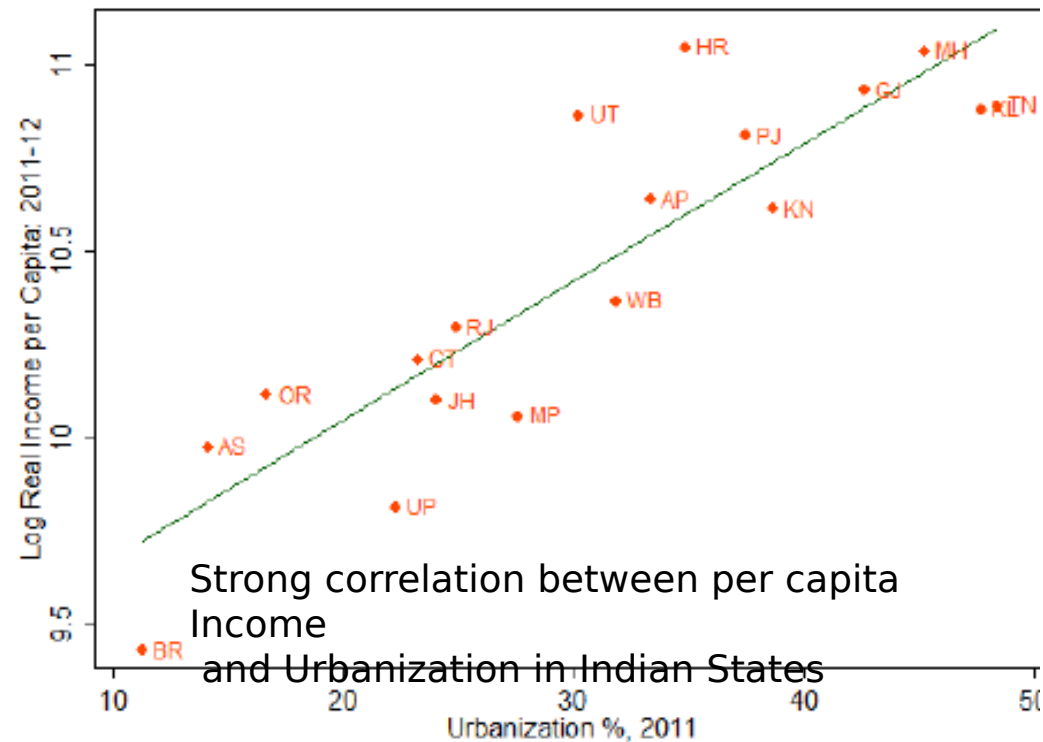
Source: Census 2001, Table B-4.

| City      | Population (Million) - 2011 | % Households with main occupation in mfg. sector - 2001 |
|-----------|-----------------------------|---------------------------------------------------------|
| Mumbai    | 18.4                        | 27                                                      |
| Delhi     | 16.3                        | 27                                                      |
| Kolkata   | 14.1                        | 27                                                      |
| Chennai   | 8.7                         | 25                                                      |
| Bengaluru | 8.5                         | 22                                                      |
| Hyderabad | 7.7                         | 18                                                      |
| Ahmedabad | 6.4                         | 32                                                      |
| Pune      | 5.0                         | 27                                                      |
| Surat     | 4.6                         | 64                                                      |
| Jaipur    | 3.1                         | 27                                                      |

# LOW LEVEL OF URBANIZATION?

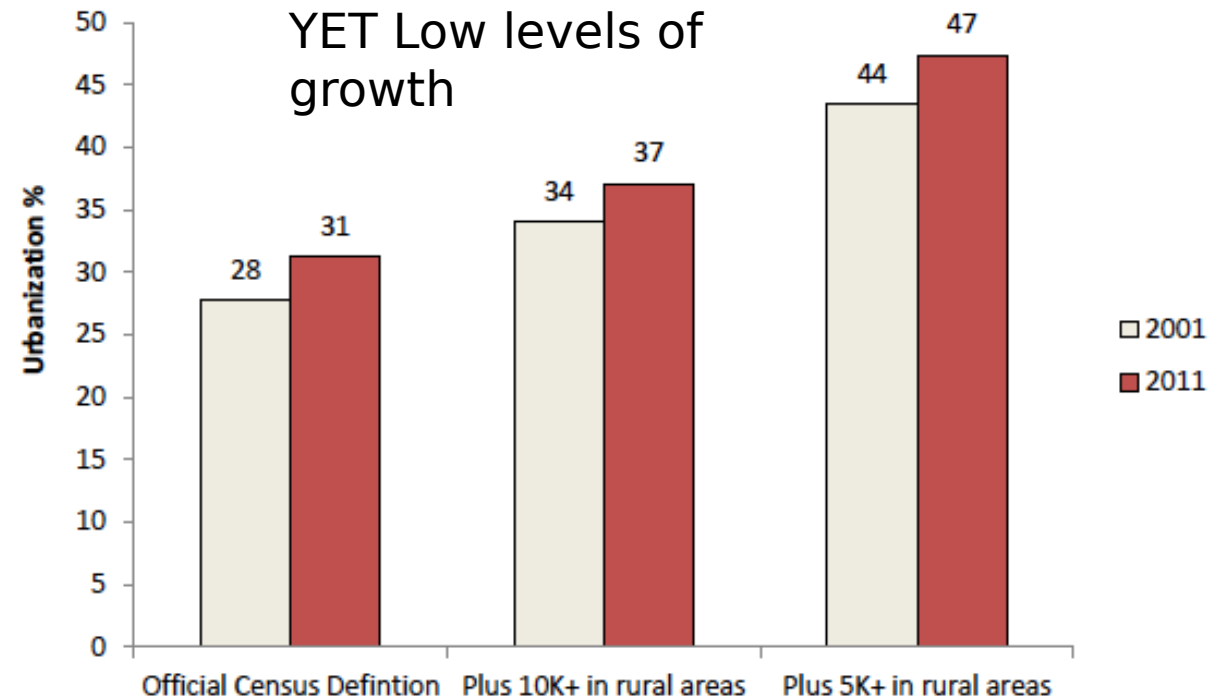
At a global level, India's urbanization rate is observed to be more than 10 percentage points lower than the rate predicted for its level of per capita income (Deshingkar & Anderson, 2004).

Figure 6 (A): Income Levels and Urbanization within India



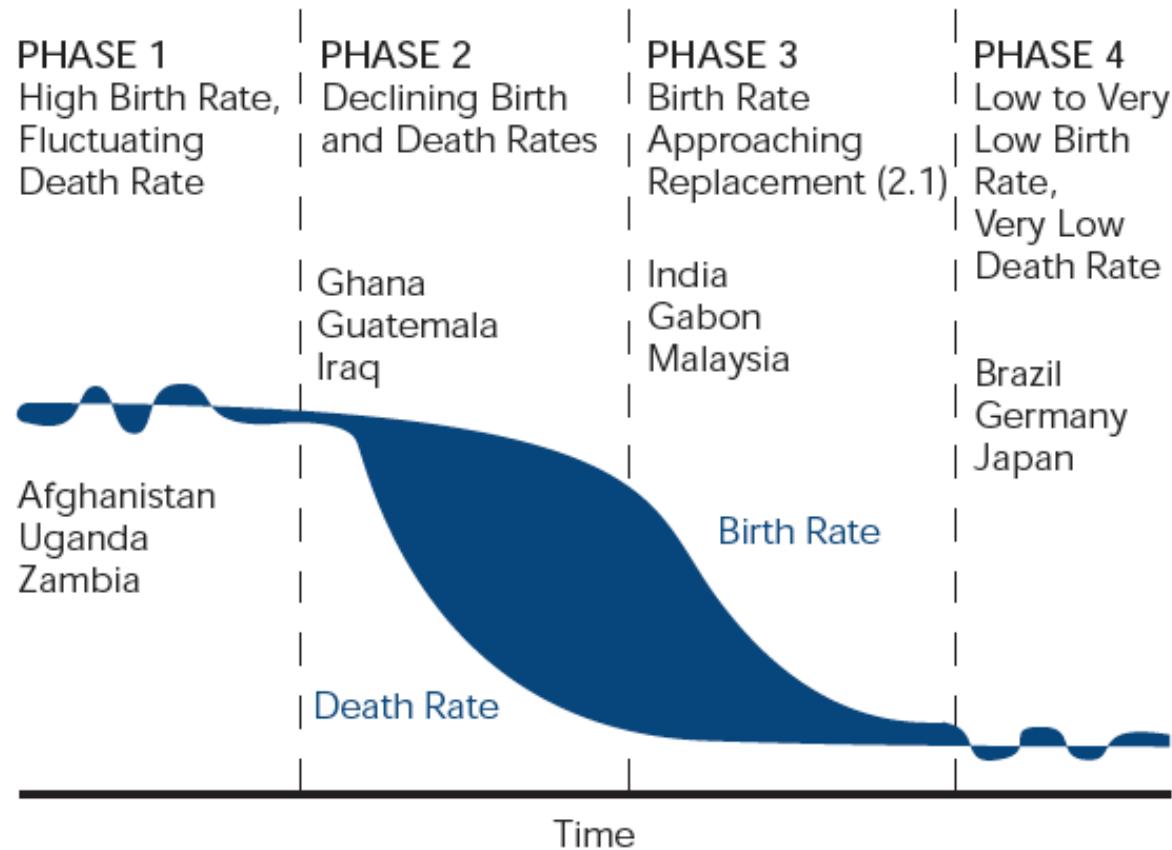
N = 18 Large States, Correlation Coefficient = +0.9

Figure 7: Urbanization in India in 2001 and 2011 across definitions



# DEMOGRAPHIC TRANSITIONS

## The Classic Phases of Demographic Transition

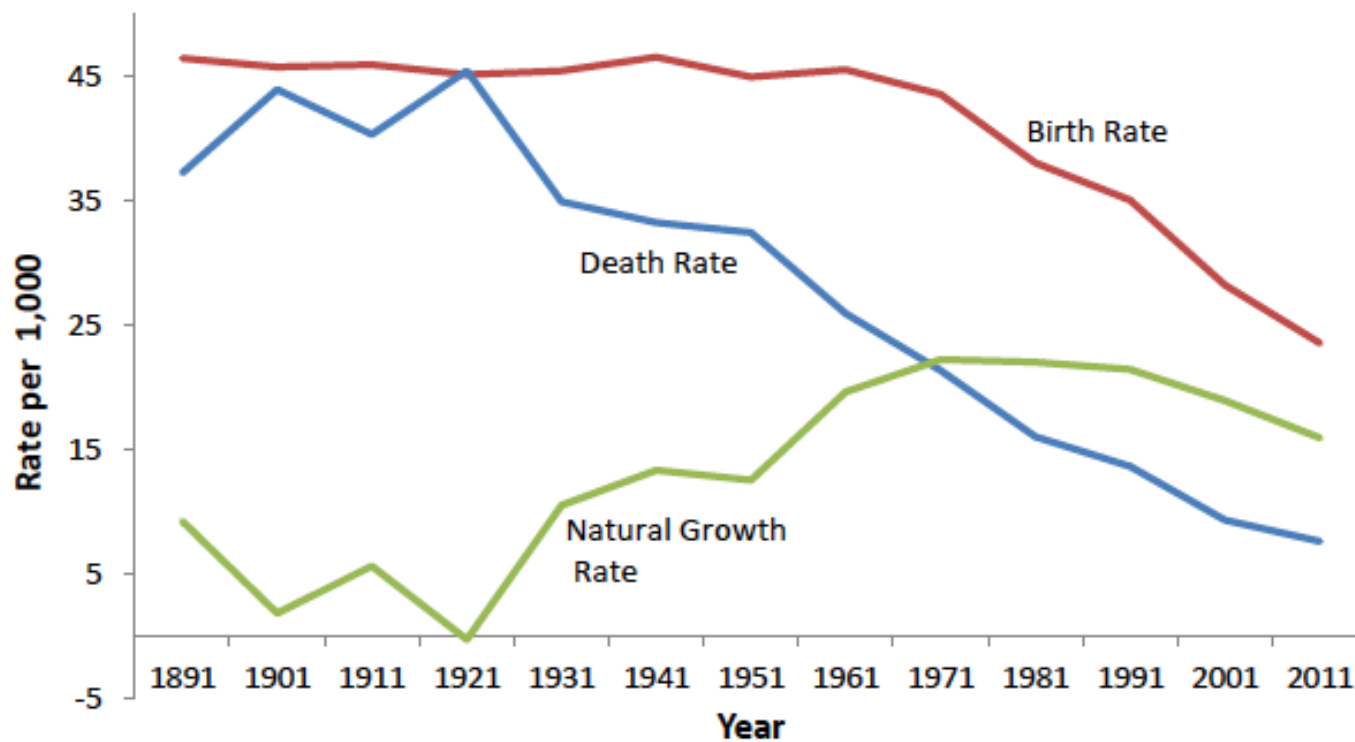


The demographic transition, on the other hand, refers to the transition from high birth and death rates to low birth and death rates

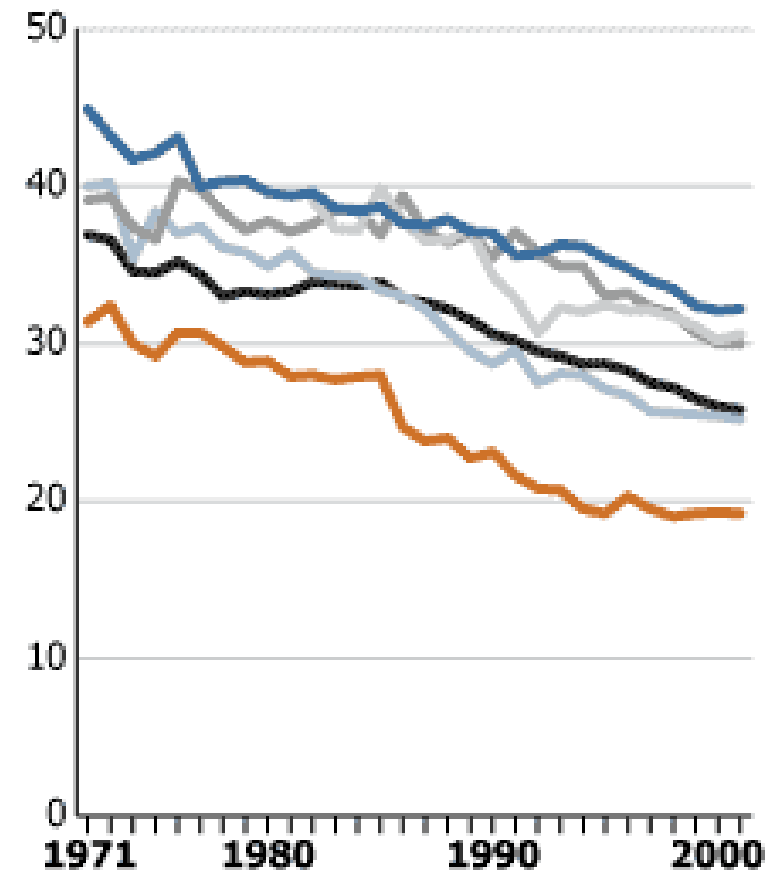
# CBR AND CDR - INDIA

Births per 1,000 population

Figure 8: The Demographic Transition of India



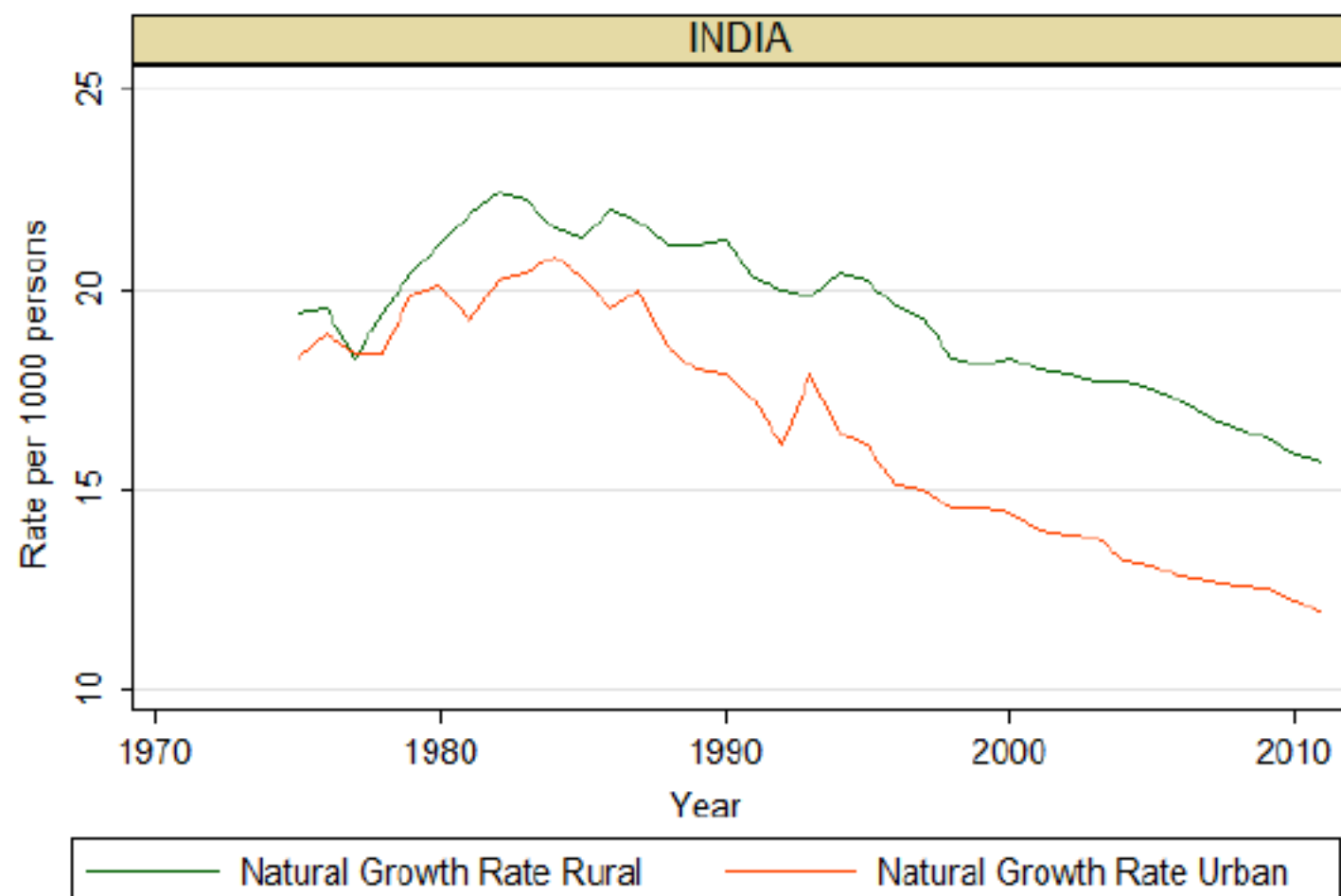
Source: Dyson (2004) and SRS statistics.



■ Uttar Pradesh\*    ■ INDIA  
 ■ Madhya Pradesh\*    ■ Gujarat  
 ■ Bihar\*    ■ Tamil Nadu



**Figure 10: Rural-Urban Demographic Divergence of Natural Growth Rates, 1975-2011**



Source: Government of India (2009) and bulletins of the Sample Registration System, 2007-2012.

# NORTH SOUTH DIVIDE

Figure 11: Demographic Divergence in the Southern States, 1975-2011

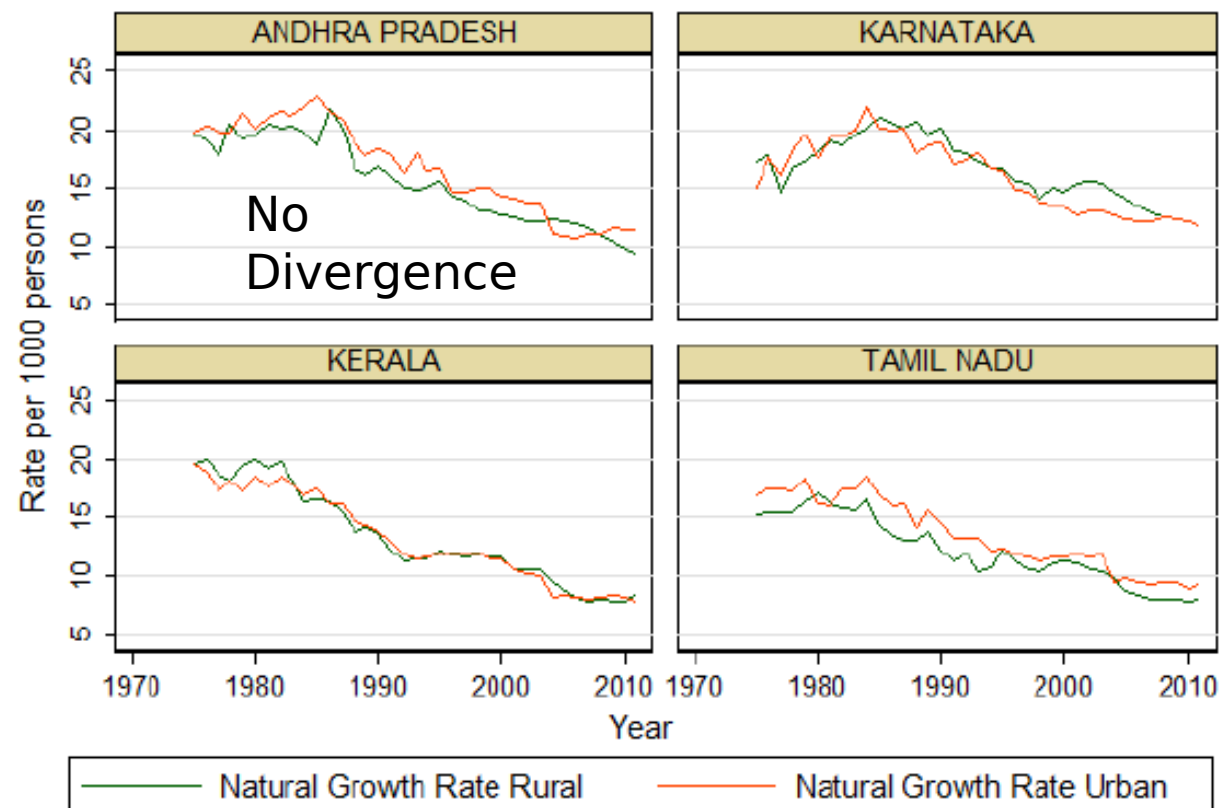
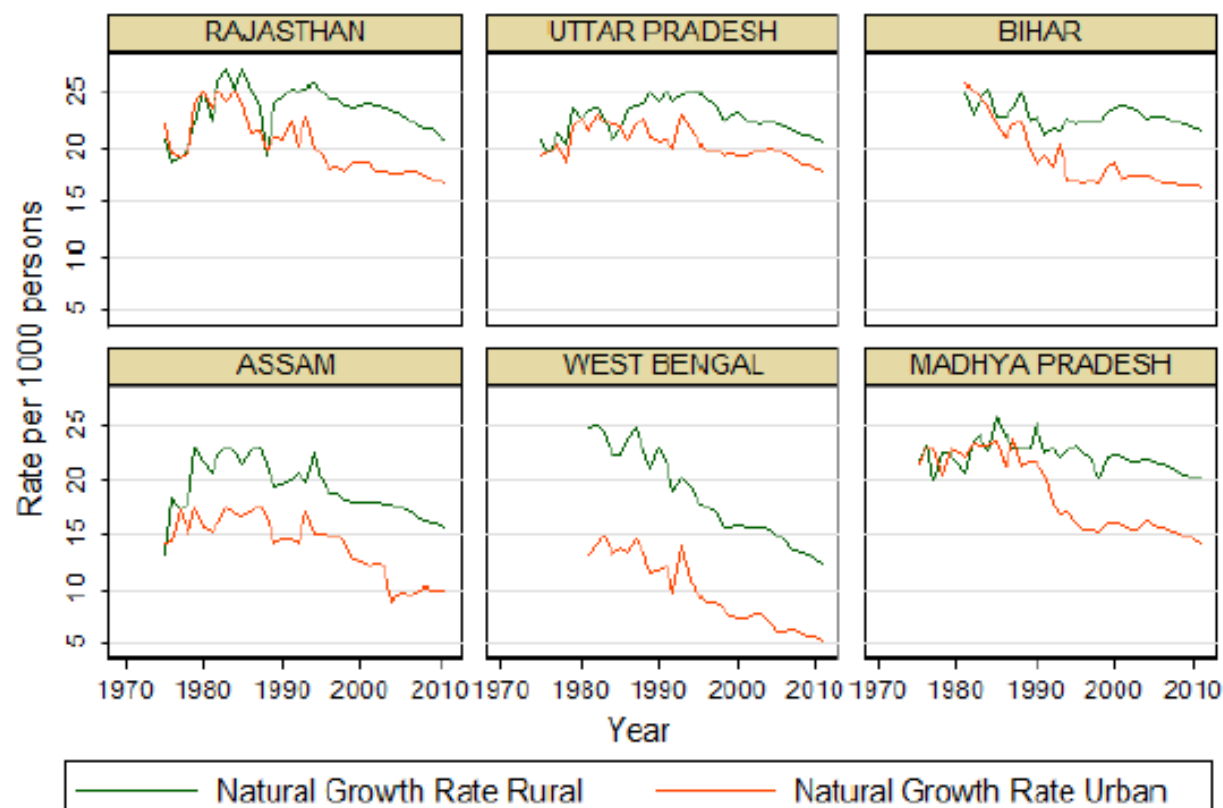
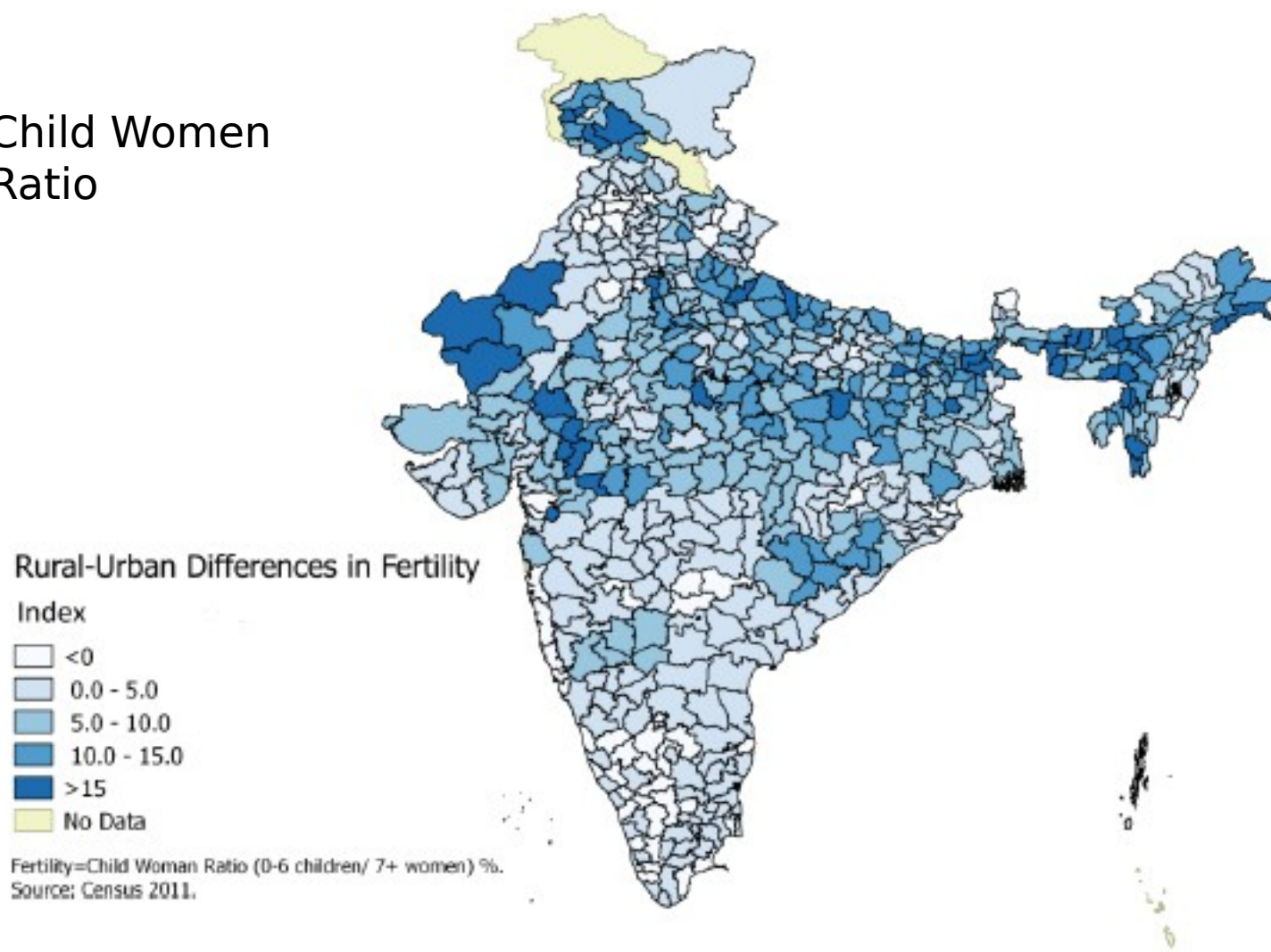


Figure 12: Demographic Divergence in Selected Northern States, 1975-2011



**Figure 13: Rural-Urban Differences in Fertility at the District Level**

Child Women  
Ratio



Source: Census 2011.

**Table 5: Impact of the Demographic Divergence on Urbanization Rates**

| Dependent Variable: Urbanization %                       |                     |                     |                     |
|----------------------------------------------------------|---------------------|---------------------|---------------------|
|                                                          | (1)                 | (2)                 | (3)                 |
| Rural-Urban Natural Growth Rate Difference (10 year lag) | -0.68<br>(0.213)*** | -0.71<br>(0.214)*** | -0.56<br>(0.202)*** |
| Urbanization % (10 year Lag)                             |                     | -0.19<br>(0.164)    | -0.12<br>(0.152)    |
| Log Number of Towns                                      |                     |                     | 6.54<br>(1.783)***  |
| Constant                                                 | 30.07<br>(1.264)*** | 34.62<br>(4.208)*** | 5.59<br>(8.79)      |
| Observations                                             | 105                 | 105                 | 105                 |
| Time (Decade) Fixed Effects                              | Yes                 | Yes                 | Yes                 |
| State Fixed Effects                                      | Yes                 | Yes                 | Yes                 |

Notes: Panel data of Indian States at 5-time period intervals: 1971, 1981, 1991, 2001, 2011. Source: Various Census volumes and SRS statistics. Standard errors reported in parentheses. \*\*\*Significant at 1% level and

\*\*Significant at 5% level.

## India's low and slow pace of urbanization thus hinges on three important factors

**Lowness:** occurs partly because of the nature of India's urban definition and

**Slowness:** occurs because of

- the *demographic divergence*
- the *highly gendered nature of work- related migration that leads to considerable return migration from cities to villages.*

Figure 14 (A): Population of large UA's in 1901 (in 2011 boundaries)

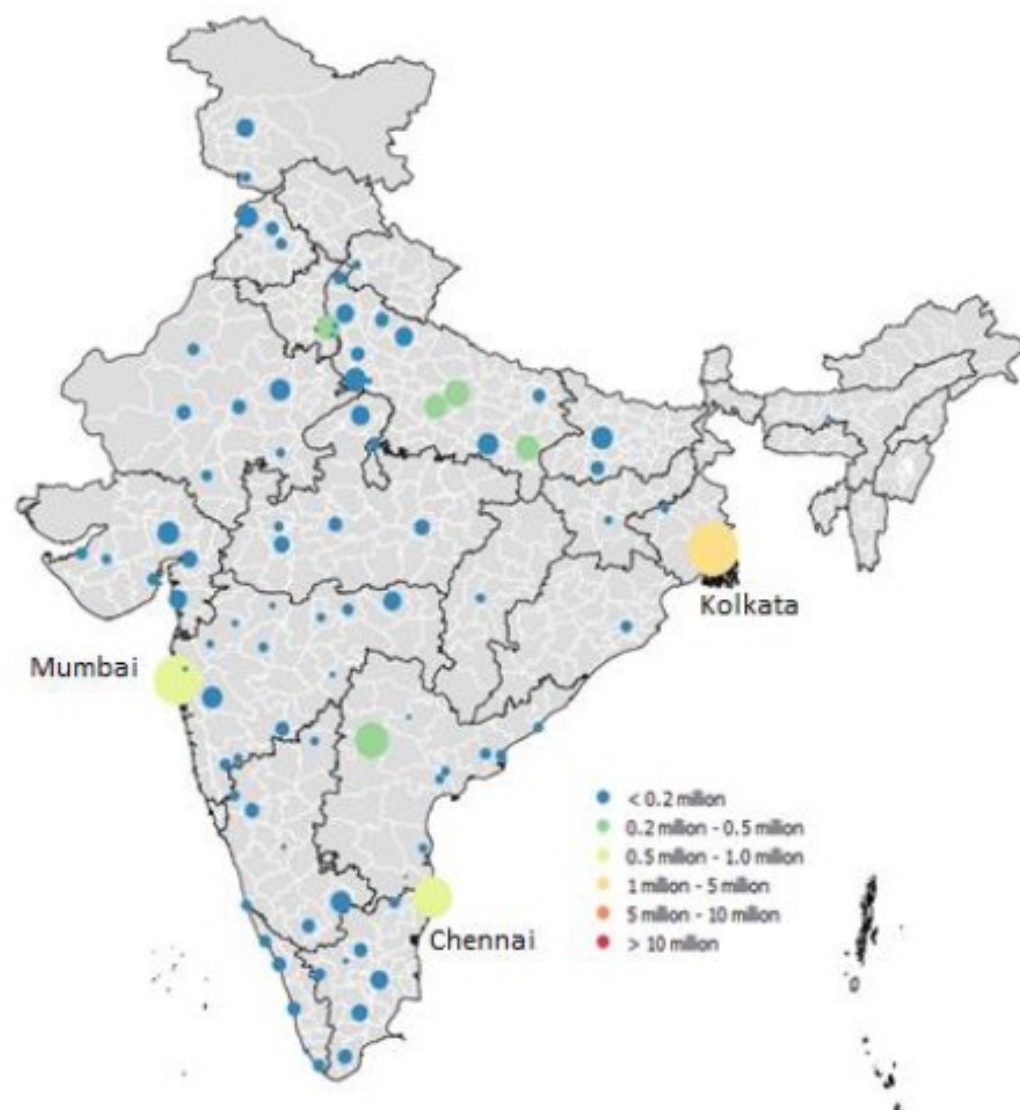
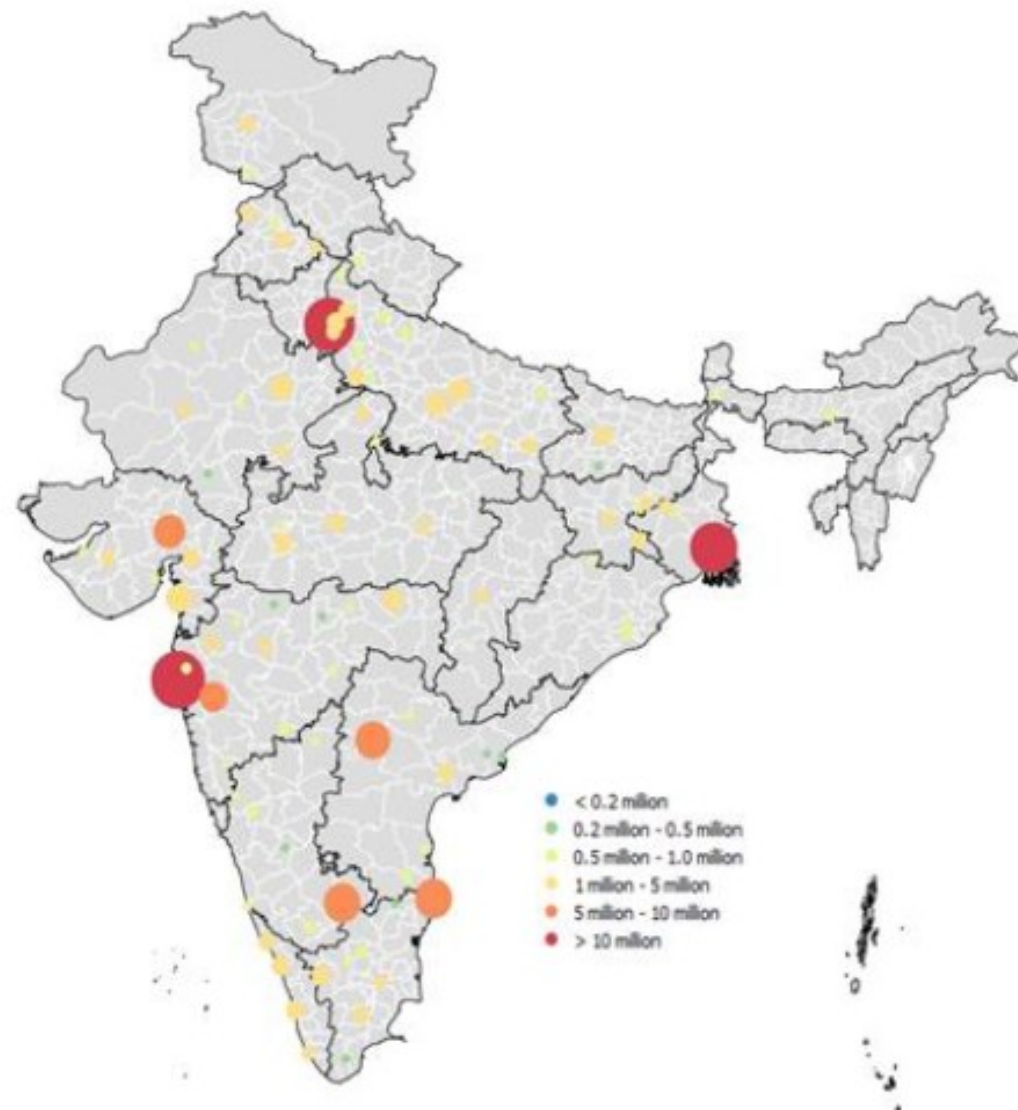


Figure 14 (B): Population of Large UA's in 2011



Source: Various Census Volumes. Top 100 Urban Agglomerations in 2001. City bubbles scaled to population size.



Thus, cities could be classified into four categories –

- **Internal growth** (with minimal migration),
- **External growth** (with minimal fertility)
- **Explosion** (with high fertility and migration)
- **Stagnation** (with low fertility and migration). The

## City Growth

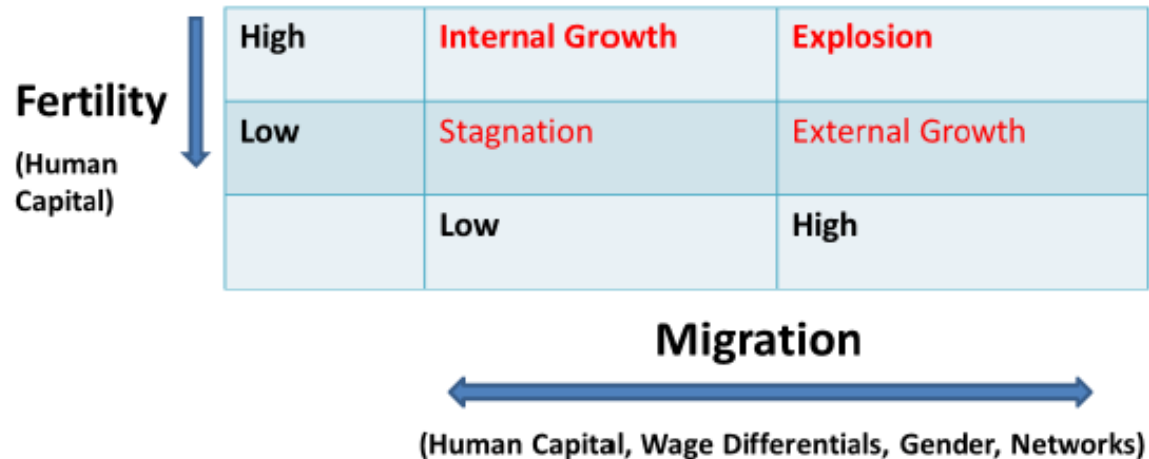
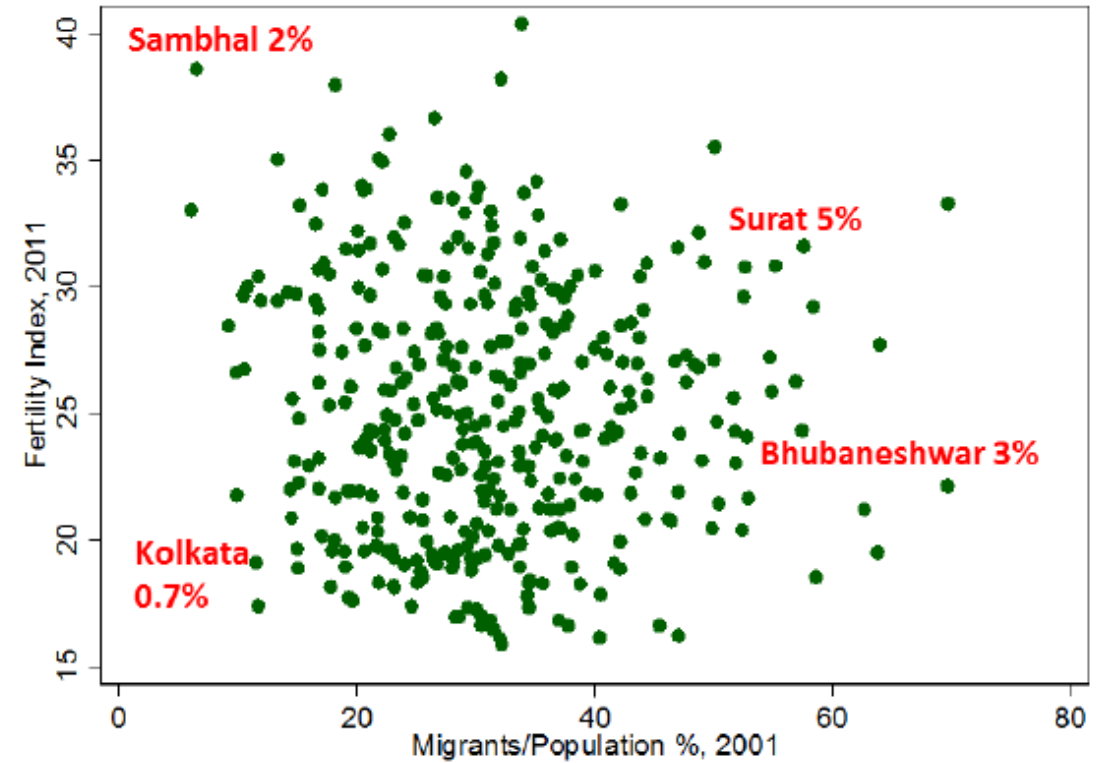


Figure 18: Fertility vs. Migration across Indian Urban Agglomerations



Source: Census 2011. N= 350+ urban agglomerations. Figures next to marked cities reflect annual growth rate between 2001 and 2011. Fertility Index = % 0-6 Age Children / Women age 7+.

# STABLE POPULATIONS AND STATIONARY POPULATION

**Stable populations** are theoretical models widely used by demographers to represent and understand the structure, growth and evolution of human populations.

- ▢ Stable populations have age-specific fertility and mortality rates that remain constant over time.
- ▢ Proved mathematically that populations with unchanging fertility and mortality patterns grow (or shrink) at a constant rate and acquire a characteristic age structure that does not change over time.

**A stationary population** is a special example of a stable population with a zero growth rate, neither growing nor shrinking in size, and is equivalent to a life table population.



# POPULATION DYNAMICS: STABLE POPULATIONS

Population renewal equation (Lotka's equation):

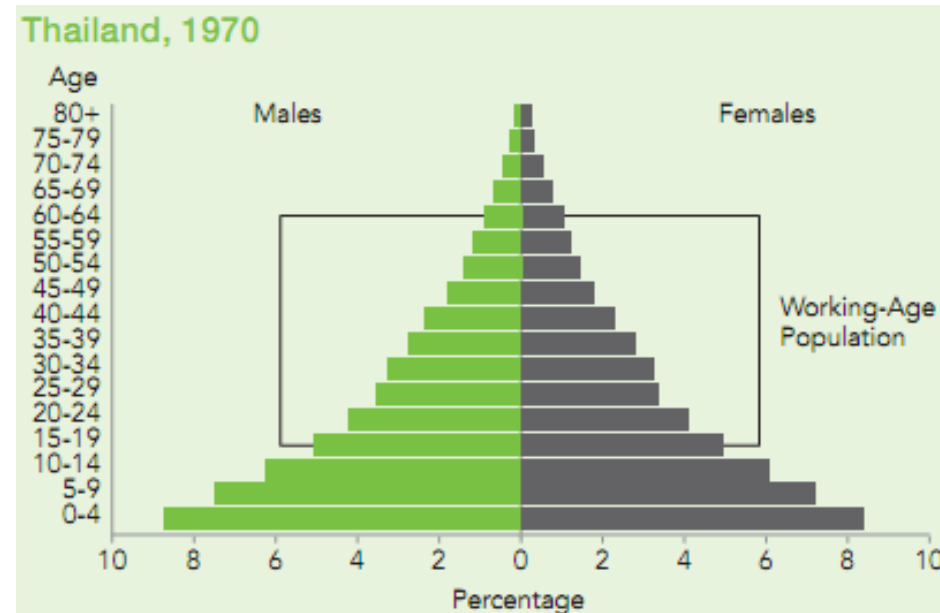
$$B(t) = \int_0^{\infty} B(t-x) p(x) \phi(x) dx$$

Births at time  $t$

$p(x)$ : Survivorship;  $\phi(x)$ : fertility at rate  $x$

$$B(t) = B_0 e^{rt}$$

Age structure:  $n(x) = n_0 e^{-rx} p(x)$



James N. Gribble and Jason Bremner, "Achieving a Demographic Dividend," Population Bulletin 67, no. 2 (2012).

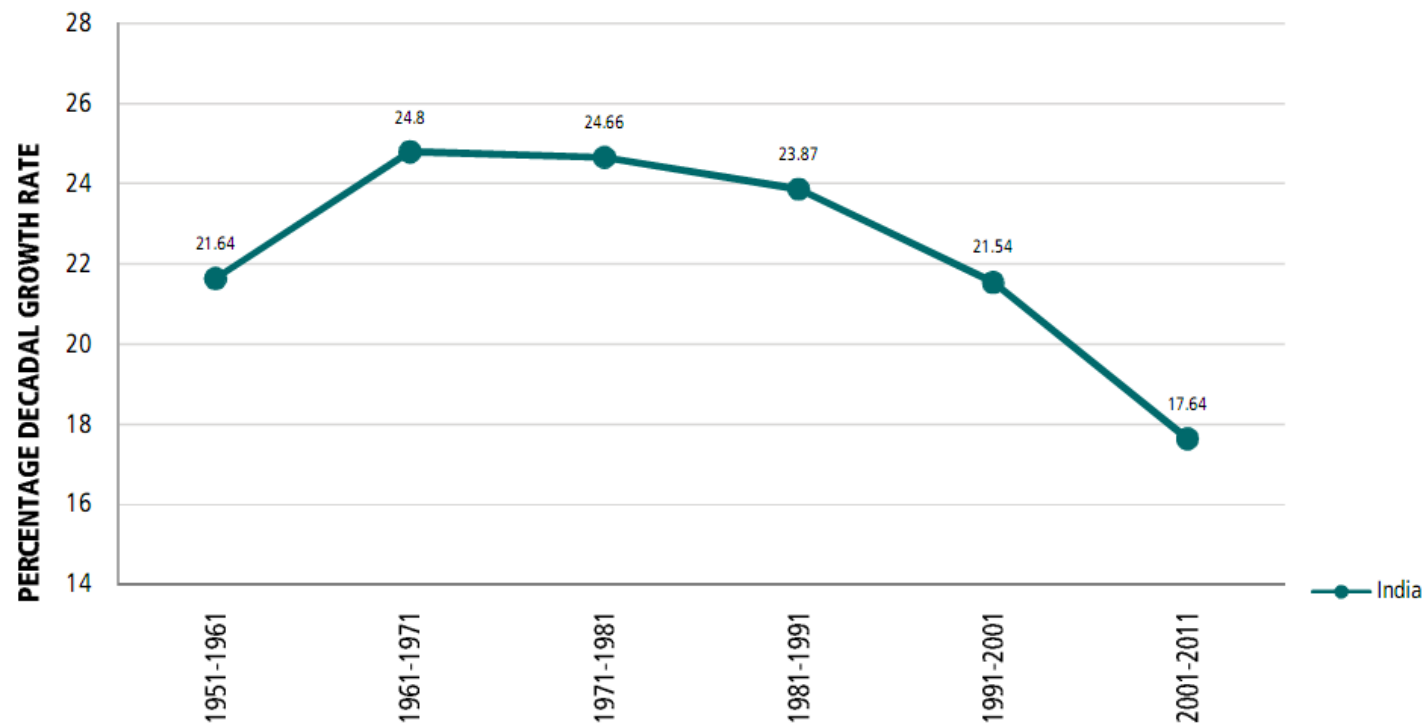
# POPULATION MOMENTUM

Current population 1 billion , TFR=3.0

Suppose TFR suddenly jumps to 2.1 (you implement some program to control)

Population growth in one year?

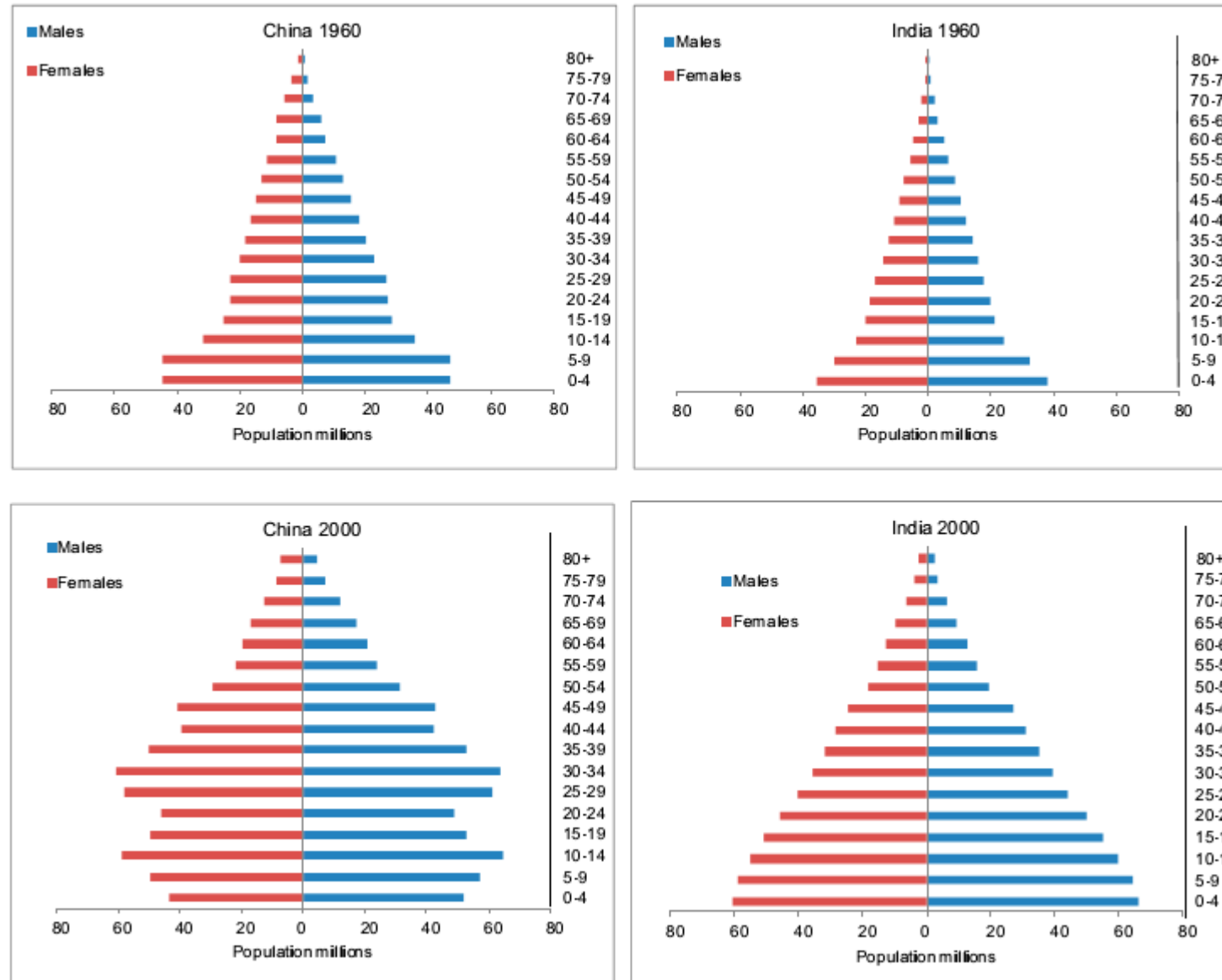
**Population momentum** refers to population growth at the national level which would occur even if levels of childbearing immediately declined to replacement level



Census of India

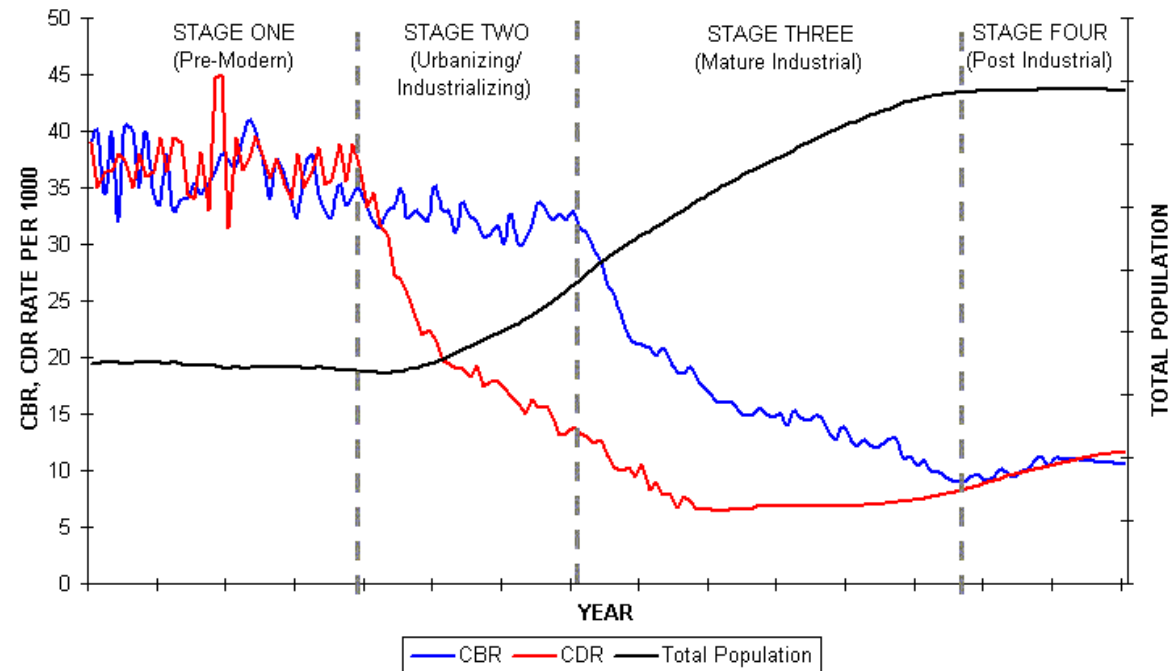
[http://censusindia.gov.in/2011-prov-results/data\\_files/india/Final\\_PPT\\_2011\\_chapter3.pdf](http://censusindia.gov.in/2011-prov-results/data_files/india/Final_PPT_2011_chapter3.pdf)

Figure 2. Comparative Evolution of Population Pyramids



# DEMOGRAPHIC TRANSITIONS

THE DEMOGRAPHIC TRANSITION MODEL



<http://www.uwmc.uwc.edu/geography/Demotrans/demtran.htm>

